Decomposing the Annuity Puzzle: A Structural Lifecycle Model and a Reduced-Form UK Transport*

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Abstract

Yaari (1965) showed that a consumer with uncertain lifetime and no bequest motive should fully annuitize under actuarially fair pricing. Observed voluntary annuity ownership among US retirees is only a few percent. I develop two complementary models that approach the gap from independent directions. **Model 1** is a structural lifecycle decomposition. A nine-channel rational-plus-preference specification (Social Security, bequests, combined medical expenditure risk and the Reichling–Smetters health-mortality correlation, subjective survival pessimism, pricing loads, inflation erosion, public-care aversion, age-varying consumption needs from Aguiar and Hurst, 2013, and state-dependent utility from Finkelstein et al., 2013) predicts ownership of 1.7% relative to a frictionless population benchmark of 41.8%. This is a slight underestimate of the two HRS empirical comparators reported in parallel—3.11% (95% CI $[2.54\%, 3.81\%]$) for the cleaner fat-file lifetime contract indicator and 5.21% (95% CI $[4.45\%, 6.09\%]$) for the conventional any-annuity income proxy used in prior literature.

*Generative AI tools were used to assist with code development, manuscript preparation, and editing. All analysis, interpretation, and conclusions are the sole responsibility of the author. Replication code and data are available at `https://github.com/DerekTharp/annuity-puzzle-model`.

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The nine-channel result does not imply that behavioral factors do not matter. In an exploratory analysis I layer two behavioral channels onto the nine-channel baseline: source-dependent utility (SDU; Blanchett and Finke, 2024, 2025) with $\lambda_W = 0.625$, and a narrow-framing at-purchase penalty (PED) consistent with Brown et al. (2008), Chalmers and Reuter (2012), and Hu and Scott (2007) with $\psi_{\text{purchase}} = 0.05$. Both parameters are exploratory best guesses anchored to literature magnitudes rather than moment-matched to US data. At these values each behavioral channel individually moves predicted ownership by more than almost any single rational, preference, or structural channel, but the two operate in opposite directions and largely offset. The eleven-channel prediction is 0.1% and is sensitive to assumptions—small shifts within the literature-defensible range produce large changes in either direction—so the nine-channel structural baseline remains the disciplined Model 1 reading. The dynamics observed in this exercise would themselves predict a meaningful amount of inconsistency in empirical and structural estimates of annuity demand across studies and over time.

Model 2 is a reduced-form transport using the UK 2015 pension-freedoms reform as out-of-sample variation. The 2015 reform shifted the regulated regime for DC pension pots from de facto compulsory annuitization—enforced pre-reform by a 55% unauthorised-payment charge on alternative withdrawals (pre-reform retention 95%)—to voluntary access with marginal-rate tax neutrality, producing post-reform voluntary retention of 17% at the mid anchor (range 13%–25% across ABI and FCA estimates). The retention ratio is applied as a behavioral wedge against the frictionless population benchmark of 41.8%, yielding predicted US ownership of 7.5% (bracket $[5.7\%, 11.0\%]$). The empirical fact that UK annuity share remained at approximately 9% in 2024/25 despite gilt yields at 14-year highs (Financial Conduct Authority, 2025) indicates that the post-2015 retention regime has been broadly stable across a full rate cycle, ruling out a substantial rate-environment confound. No US moment is used to discipline Model 2; consistency with the HRS comparators is an out-of-sample comparison. The conventional HRS income proxy lies inside the Model 2 bracket; the cleaner lifetime-

contract indicator sits just below it.

The two models reach the same neighborhood from independent directions: structural decomposition from inside the lifecycle problem, and cross-country transport from outside it. Channel attribution within Model 1 uses exact Shapley values computed over all $2^{11} = 2{,}048$ subsets of the eleven channels.

JEL Codes: D14, D15, D91, G22, H55, J14

Keywords: annuity puzzle, lifecycle model, Shapley decomposition, stochastic mortality, age-varying consumption needs

1 Introduction

Yaari (1965) established one of the sharpest predictions in consumer theory: an individual facing uncertain lifetime, with no bequest motive and access to actuarially fair annuities, should convert all wealth into a life annuity. The mortality credit—the return premium financed by redistributing wealth from decedents to survivors—dominates any asset with the same underlying return. Davidoff et al. (2005) extended this result to incomplete markets, showing that partial annuitization remains optimal under a wide range of conditions. Yet observed voluntary annuity ownership among single US retirees aged 65–69 is only a few percent: 3.11% on the cleaner HRS lifetime annuity contract indicator (q286 series) and 5.21% on the conventional any-annuity income proxy used in prior literature (Lockwood, 2012; Dushi and Webb, 2004; Poterba et al., 2011).[1] Six decades of research have proposed partial explanations for this gap, but no single paper has nested all the major channels in a unified framework and measured their joint quantitative effect.

This paper provides a calibrated structural accounting exercise for that gap, supplemented by an external reduced-form transport. Model 1 is a structural lifecycle decomposition with nine rational-plus-preference channels: pre-existing Social Security annuitization, bequest motives, combined medical expenditure risk and the health-mortality correlation identified by Reichling and Smetters (2015), subjective survival pessimism (O'Dea and Sturrock, 2023), pricing loads, inflation erosion, public-care aversion (Ameriks et al., 2020), age-varying consumption needs (Aguiar and Hurst, 2013), and state-dependent utility (Finkelstein et al., 2013). Together these nine channels predict ownership of 1.7%, a slight underestimate of the two HRS empirical comparators (3.11% and 5.21%). The structural model accounts for most of the gap through standard mechanisms, leaving a small residual relative to the conventional measure; I do not claim it "matches" or "resolves" the puzzle.

[1] The two HRS measures differ because the income proxy (*rwiann*) classifies any positive annuity income as ownership, including one-time DC pension withdrawals and short-period payouts; the lifetime contract indicator (the question stem "Will this continue for the rest of your life?") isolates true life-contingent annuity contracts. Section 3 discusses both measures in detail.

Model 1 also reports an eleven-channel extended specification that layers two behavioral channels onto the nine-channel baseline: source-dependent utility (SDU; Shefrin and Thaler, 1988; Thaler, 1999; Blanchett and Finke, 2024, 2025) operationalized as a $\lambda_W = 0.625$ discount on portfolio drawdowns, and a narrow-framing at-purchase penalty (PED) consistent with Tversky and Kahneman (1992), Hu and Scott (2007), Brown et al. (2008), and Chalmers and Reuter (2012) parameterized as $\psi_{\text{purchase}} = 0.05$. Both parameters are exploratory best guesses anchored to literature magnitudes; they are *not* moment-matched to a US identification target. The eleven-channel specification predicts 0.1% because the at-purchase penalty saturates at the chosen literature magnitude; this overshoots the empirical comparators in the opposite direction. The nine-channel baseline is therefore the disciplined Model 1 reading; the eleven-channel specification is reported as a literature-magnitude exercise that shows where standard behavioral parameters would push predictions if introduced in isolation.

Model 2 is a reduced-form transport that bypasses internal behavioral parameterization. The UK 2015 pension-freedoms reform shifted regulated DC pension-pot decumulation from de facto compulsory annuitization (enforced by a 55% unauthorised-payment charge on alternative withdrawals; pre-reform retention 95%) to voluntary access with marginal-rate tax neutrality. Post-reform voluntary retention is 17% at the mid anchor (range 13%–25% across ABI and FCA estimates). The post-to-pre retention ratio is applied as a multiplicative behavioral wedge against the frictionless population benchmark of 41.8%, yielding predicted US ownership of 7.5% (bracket $[5.7\%, 11.0\%]$). FCA Retirement Income Market Data (Financial Conduct Authority, 2025) report annuity share at approximately 9% in 2024/25 despite gilt yields at 14-year highs, indicating that the post-2015 retention regime has been broadly stable across a full rate cycle. The English Longitudinal Study of Ageing (ELSA) pension grid provides supporting individual-level evidence: 90.2% of wave 6 (2012–13) DC pension recipients reported annuity-style income under the pre-freedoms mandate, and pooling waves 8–11 (2016–24), 1.27% of observed lump-sum disposition records used the lump sum to buy annuity income (n=869). No US moment is used to discipline Model 2; consistency with the

HRS comparators is an out-of-sample comparison.

The paper's contribution is fourfold. First, the unified structural framework—the first to nest age-varying consumption needs alongside the established rational and preference channels of the annuity-puzzle literature. Second, the cross-validation of a structural decomposition against an external reduced-form transport that anchors aggregate behavioral attrition to a single market-design moment. Third, an exact Shapley decomposition over all $2^9 = 512$ subsets of the nine structural channels (the disciplined headline attribution), with an exploratory extension to $2^{11} = 2{,}048$ subsets that adds the two literature-magnitude behavioral parameters. Fourth, two distinct policy levers emerge from the comparison: a supply-side reform (group pricing) and a demand-side reform (default architecture) operate on different margins.

The results have a three-layer structure. First, under modern annuity pricing (MWR = 0.87, Wettstein et al., 2021), the six rational channels alone predict ownership of 8.5% relative to a frictionless population benchmark of 41.8%; adding the two preference channels brings the prediction to 1.8%; adding public-care aversion (χ_{LTC}) yields the nine-channel structural baseline of 1.7%. Second, the eleven-channel extended specification layers SDU and PED at literature-magnitude parameters and predicts 0.1%; the at-purchase penalty saturates and the model overshoots in the opposite direction, illustrating that small-parameter behavioral additions can move predictions discontinuously and motivating the disciplined nine-channel reading. Third, Model 2 applies the UK 2015 post/pre retention ratio to the frictionless benchmark and predicts 7.5% (bracket $[5.7\%, 11.0\%]$). Both HRS empirical comparators (5.21% for the conventional income proxy; 3.11% for the cleaner lifetime-contract indicator) are reported as out-of-sample comparisons against both models.

Relative to prior multi-channel papers, the present model broadens the channel set and changes the attribution. Pashchenko (2013) jointly considered Social Security, bequests, minimum purchase requirements, and pricing frictions, but not the Reichling–Smetters health-survival mechanism, subjective survival pessimism, or age-varying needs. Peijnenburg et al.

(2016) emphasized incomplete markets and background risk, concluding that the annuity puzzle remains unresolved in their framework. This paper shows that pricing, correlated health and mortality risk, and late-life expenditure decline are quantitatively central once evaluated in a common model.

The nine-channel structural Shapley over all $2^9 = 512$ subsets of the structural channels provides the paper's preferred order-independent attribution. Among demand-suppressing channels, pricing loads (28.6 pp, 71%) and the combined medical risk and Reichling–Smetters health-mortality correlation channel (10.3 pp, 26%) are the two largest contributors at the structural baseline. Social Security enters with a negative Shapley value (-26.3 pp, -65%), reflecting its role as a complement that raises demand at the margin. State-dependent utility, bequest motives, and public-care aversion contribute small magnitudes in the current enumeration.[2] An eleven-channel exploratory extension that adds SDU ($\lambda_W = 0.625$) and PED ($\psi_{\text{purchase}} = 0.05$) yields large opposite-sign behavioral contributions; under the literature-magnitude parameterization the PED contribution dominates, but the behavioral attributions are illustrative rather than identified.

The policy and welfare results extend beyond the ownership match. Group annuity pricing (MWR = 0.90) raises predicted ownership relative to the production calibration. Removing the at-purchase penalty ($\psi_{\text{purchase}} = 0$) recovers a counterfactual broadly consistent with the UK 2015 pension-freedoms reform and with the Goda et al. (2014) federal Thrift Savings Plan default-architecture evidence; neither is a clean structural identification of the underlying mechanism. Social Security cut counterfactuals reveal that a 23% benefit cut—the projected consequence of trust fund exhaustion around 2033—raises private annuity demand substantially, but the largest cuts eventually reduce demand by removing the income floor that makes private annuitization feasible at all. Under current pricing the welfare gain from annuity market access is small across the wealth distribution because most

[2] The public-care aversion (LTC) Shapley value reported here is computed from the existing subset enumeration. The next pipeline run incorporates a corrected consumption-equivalent treatment of χ_{LTC} that is expected to raise its contribution; the prose ordering above reflects the structural channels whose ranking is robust across that correction.

calibrated households optimally do not annuitize, so access is approximately neutral; policy interventions that change the optimal choice (group pricing, demand-side architecture) generate the larger welfare gains.

These results shift the question away from whether retirees irrationally fail to annuitize and toward a more policy-relevant one: which channels suppress annuity demand most, and which market reforms would make annuitization welfare-improving for particular households?

Section 2 specifies the model. Section 3 describes the calibration. Section 4 presents the decomposition, counterfactuals, and welfare analysis. Section 5 reports sensitivity checks. Section 6 discusses limitations. Section 7 concludes.

2 Model

2.1 Environment

Time is discrete. An individual enters at age 65 (period $t = 1$) and may survive to a maximum age of 110 (period $T = 46$). At $t = 1$, the individual makes a single irreversible decision: what fraction $\alpha \in [0, 1]$ of initial liquid wealth W_0 to convert into a single premium immediate annuity (SPIA). The remaining wealth $(1-\alpha)W_0$ stays liquid. In each subsequent period, the individual chooses consumption c_t from available resources.

The state at each period t is characterized by four variables: liquid wealth $W_t \geq 0$, annuity income A (fixed after the age-65 purchase), health status $H_t \in \{1, 2, 3\}$ (Good, Fair, Poor), and age t. Social Security income $\mathrm{SS}(t)$ is received exogenously in each period and is not a decision variable.

2.2 Preferences

The individual maximizes expected discounted lifetime utility. Flow utility over consumption takes the constant relative risk aversion (CRRA) form:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1. \tag{1}$$

The coefficient of relative risk aversion γ governs both risk aversion and the elasticity of intertemporal substitution.

Bequest utility follows the De Nardi–French–Jones specification, which nests bequests as a luxury good:

$$v(b) = \theta \frac{(b+\kappa)^{1-\gamma}}{1-\gamma}, \tag{2}$$

where b is bequeathable wealth (liquid wealth at death; annuity income ceases), $\theta \geq 0$ controls bequest intensity, and $\kappa > 0$ is a shifter that determines the degree to which bequests are a luxury good (De Nardi, 2004). When κ is large relative to typical wealth levels, the marginal utility of bequests is nearly flat for low-wealth individuals and declines steeply for the wealthy. This specification, estimated by Lockwood (2012) using Health and Retirement Study (HRS) data, captures the empirical pattern that bequest motives are concentrated among high-wealth households.

The parameters θ and κ must be interpreted jointly. With the DFJ luxury-good calibration ($\theta = 56.96$, $\kappa = \$272{,}628$), an individual with \$50,000 in wealth faces negligible marginal bequest utility, while an individual with \$1,000,000 faces strong resistance to annuitizing. Combining this θ with a small κ (e.g., \$10, the homothetic specification) produces pathological results: extreme marginal bequest utility near $b = 0$ acts as a precautionary buffer rather than a genuine bequest motive, anomalously *increasing* predicted ownership above the no-bequest case.[3] The DFJ estimates from Lockwood (2012) are jointly calibrated to

[3] At $\kappa = \$10$, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-\gamma}$, which is extreme relative to marginal consumption utility. Individuals avoid annuitizing not to protect bequests, but to avoid near-zero-wealth states where $v'(b)$ becomes very large. The implementation applies a small numerical floor at $\max(b+\kappa, 1)$

HRS bequest data; using one parameter with the other replaced is not a valid counterfactual.

The bequest channel's isolated effect on ownership is modest under the DFJ specification (100% retention rate in the sequential decomposition) because the luxury-good κ makes the bequest motive nearly irrelevant for the median retiree in the HRS sample. The 100% retention rate in the sequential decomposition reflects the ordering: bequests are added when ownership is already 100% (steps 0–1 include only SS). In the full nine-channel structural model, removing bequests raises ownership from 1.7% to 21.0% (Table 7), because bequests interact with pricing loads and inflation to suppress demand for agents near the annuitization margin. The Shapley decomposition, which averages over all orderings, assigns bequests a value of 0.7 pp (2% share).

The individual discounts future utility at rate β per period.

2.3 Health dynamics and mortality

Health follows a first-order Markov process with three states: Good ($H = 1$), Fair ($H = 2$), and Poor ($H = 3$). Transition probabilities are age-dependent and calibrated to HRS panel data following De Nardi et al. (2010):

$$\Pr(H_{t+1} = h' \mid H_t = h, \text{age} = t) = \pi(h, h', t). \tag{3}$$

Mortality depends on health through a proportional hazard specification. Baseline survival probabilities $s_{\text{base}}(t)$ come from the SSA period life table used by Lockwood (2012). Health adjusts survival through:

$$s(t, H) = s_{\text{base}}(t)^{\mu(H,t)}, \tag{4}$$

where $\mu(H, t)$ is a health- and age-specific hazard multiplier with $\mu(\text{Good}, t) < 1 < \mu(\text{Poor}, t)$

to prevent overflow at $b = 0$ when $\kappa = 0$ is used as a robustness check; with the production DFJ calibration (κ = \$272,628) the floor never binds. See Appendix C for details.

and $\mu(\text{Fair}, t) = 1$. This specification maps directly to proportional hazards: if the baseline hazard is $\lambda_{\text{base}}(t) = -\ln s_{\text{base}}(t)$, then the adjusted hazard is $\mu(H, t) \cdot \lambda_{\text{base}}(t)$.

The baseline specification uses constant hazard multipliers $\mu = [0.50, 1.0, 3.8]$ for Good, Fair, and Poor health respectively, chosen as a central tendency across age bands from the HRS data. The health-mortality gradient compresses with age: the Poor-to-Fair hazard ratio falls from 3.29 at ages 65–74 to 2.77 at 75–84 to 1.82 at 85+. Section 5 reports results across the empirical range of multiplier specifications.

The age-varying multipliers are the mechanism underlying the Reichling and Smetters (2015) result. A negative health shock simultaneously increases mortality ($\mu(\text{Poor}, t) > 1$ reduces expected remaining annuity payments) and, through the medical expenditure process described below, increases the need for liquid wealth. The annuity becomes "valuation-risky": it loses value in precisely the states where liquidity is most valuable. The effect is strongest at younger ages where the gradient is widest, which matters because the annuitization decision is made at age 65.

2.4 Subjective survival beliefs

O'Dea and Sturrock (2023) document that individuals aged 65–69 underestimate their survival probability at age 75 by approximately 15 percentage points relative to actuarial tables (subjective Pr(survive to 75) ≈ 0.71 versus actuarial 0.86). This pessimism reduces the perceived value of annuities by overweighting the probability of dying before recovering the premium.

The model allows the agent's subjective survival probabilities to differ from the objective rates used by the insurer to price annuities:

$$s^{\text{subj}}(t, H) = \psi \cdot s(t, H), \quad \psi \in (0, 1]. \tag{5}$$

The parameter ψ scales each one-year survival probability downward. The annuity is priced using the objective survival rates $s(t, H)$; only the agent's Bellman equation uses s^{subj}. A per-year factor of $\psi = 0.960$ matches the O'Dea–Sturrock finding: over a 10-year horizon, $(0.71/0.86)^{1/10} \approx 0.960$. When $\psi = 1$, beliefs are objective and this channel is inactive.

2.5 Medical expenditures

Out-of-pocket medical expenditures depend on age and health. Log medical spending follows:

$$\ln m_t = \mu_m(t, H_t) + \sigma_m(t, H_t) \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \tag{6}$$

where μ_m and σ_m are the mean and standard deviation of log expenditures, both age- and health-dependent. The baseline calibration matches the age profiles reported by Jones et al. (2018): mean out-of-pocket spending of \$4,200 at age 70, rising to \$29,700 at age 100, with a 95th percentile of approximately \$111,200 at age 100.

A Medicaid safety net provides a consumption floor $\underline{c}$. If total resources (wealth plus income minus medical costs) fall below $\underline{c}$, the government covers the shortfall. The individual consumes $\underline{c}$ and enters the next period with zero liquid wealth. This floor is calibrated to the SSI federal benefit rate plus average state supplement for an elderly individual (\$6,180, following the value used in Lockwood, 2012).

Expectations over medical expenditure shocks are computed using nine-node Gauss–Hermite quadrature. At the calibrated variance ($\sigma \approx 1.4$), lower-order rules place insufficient weight in the tails of the lognormal distribution; Appendix G documents the convergence diagnostics.

2.6 Annuity pricing

The annuity is nominal: the insurer prices the stream of nominal payments using the nominal discount rate $r_{\text{nom}} = (1 + r)(1 + \pi) - 1$, where r is the real interest rate and π is the inflation rate. The actuarially fair annual payout per dollar of premium is:

$$\text{payout}_{\text{fair}} = \frac{1}{\sum_{k=0}^{T-1} \frac{\prod_{j=0}^{k-1} s_{\text{base}}(j)}{(1+r_{\text{nom}})^k}}. \tag{7}$$

The nominal discount rate reflects the insurer's investment in nominal bonds. The resulting initial payout is higher than under real discounting, but the consumer's real income from the annuity declines each period:

$$A_t^{\text{real}} = \frac{A_1}{(1+\pi)^{t-1}}, \tag{8}$$

where π is the annual inflation rate. At $\pi = 2\%$, purchasing power falls by 39% over 25 years.

When $\pi = 0$ (the Yaari benchmark and intermediate decomposition steps where inflation is inactive), $r_{\text{nom}} = r$ and the pricing reduces to the standard real-annuity formula.

The annuity is priced with a money's worth ratio (MWR), defined as the expected present value of payouts per dollar of premium. The loaded payout rate is MWR $\times$ payout$_{\text{fair}}$. I set MWR $= 0.87$ at baseline, consistent with the population-table estimates of Mitchell et al. (1999) for the US individual annuity market. This choice merits discussion. Mitchell et al. (1999) estimated MWRs of 0.80–0.85 using population mortality tables, which is the relevant benchmark for an individual considering annuitization (the "money's worth" of the purchase against population life expectancy). Using annuitant mortality tables—which adjust for selection—yields higher MWRs of 0.90–0.94, but this overstates the value proposition for the marginal non-annuitant. More recent estimates from Wettstein et al. (2021) suggest some improvement, with population-table MWRs around 0.85. Section 5 reports full sensitivity: at MWR $= 0.85$, predicted ownership rises to 0.4%; at MWR $= 0.90$, to 5.8%. The steep sensitivity to MWR is itself a finding—it identifies pricing as the most actionable policy lever.

A fixed cost of $2,500 is incurred upon any purchase.[4]

2.7 Budget constraint

Within each period, the timing is:

1. The individual enters with liquid wealth W_t, annuity income A_t, Social Security $\mathrm{SS}(t)$, and health H_t.
2. Medical expenditure m_t is realized.
3. The individual chooses consumption c_t.
4. Remaining wealth earns the risk-free return r.
5. Health transitions to H_{t+1}.
6. The individual survives to $t+1$ with probability $s(t, H_t)$.

The intertemporal budget constraint is:

$$W_{t+1} = (1+r)(W_t + A_t + \mathrm{SS}(t) - m_t - c_t), \tag{9}$$

subject to $c_t \geq \underline{c}$ and $W_{t+1} \geq 0$ (no borrowing).

2.8 Age-varying consumption needs

Aguiar and Hurst (2013) document that consumption expenditure declines at roughly 2% per year in retirement, even among wealthy households, once time costs and home production are accounted for. This decline reflects changing consumption needs—reduced transportation, clothing, and food-away-from-home expenditures—rather than binding liquidity constraints. The declining profile reduces the value of late-life income streams, including annuities.

[4] The MWR is computed using beginning-of-period payment timing: the first payment occurs at purchase. This convention is standard in the annuity pricing literature (Mitchell et al., 1999) and matches the implementation in the replication code.

The model captures this through an age-dependent weight on flow utility:

$$w(t) = (1 - \delta_c)^{t-1}, \quad \delta_c = 0.02, \tag{10}$$

where $t = 1$ at age 65. Flow utility becomes $w(t) \cdot u(c)$. This specification is equivalent to an age-dependent felicity shifter: as consumption needs decline, the marginal value of additional consumption in late life falls, reducing the insurance value of an annuity that provides a level real income stream. When $\delta_c = 0$, this channel is inactive and the model reduces to the standard time-separable case.

Age-varying consumption needs should be distinguished from time preference (β). The discount factor β governs the rate at which future utility is discounted; $w(t)$ governs how much utility a given level of consumption generates at each age. A constant β with declining $w(t)$ implies that the individual values future consumption opportunities but needs less consumption to achieve the same felicity as she ages.

2.9 State-dependent utility

Finkelstein et al. (2013) estimate that the marginal utility of consumption declines with poor health, with a central estimate implying that individuals in poor health value a dollar of consumption roughly 10–25% less than those in good health. Reichling and Smetters (2015) incorporated this channel in their annuity model. State-dependent utility reduces annuity demand because the annuity delivers income in poor-health states where its marginal utility is lower.

The model allows the marginal utility of consumption to depend on health status through a multiplicative weight:

$$u(c, H) = \varphi(H) \cdot \frac{c^{1-\gamma}}{1-\gamma}, \tag{11}$$

where $\varphi(\text{Good}) = 1.00$, $\varphi(\text{Fair}) = 0.92$, and $\varphi(\text{Poor}) = 0.82$, following the mapping used by Reichling and Smetters (2015) based on the estimates of Finkelstein et al. (2013). When

$\varphi(H) = 1$ for all H, this channel is inactive.

In practice, state-dependent utility has a negligible effect on predicted ownership (Shapley value of 0.3 pp). The channel is included for completeness and to confirm that its quantitative irrelevance, suggested by the prior literature, holds in the unified framework.

2.10 Behavioral channels (exploratory) and the UK reduced-form transport

The nine rational and preference-based channels above constitute the disciplined Model 1 baseline. Two further behavioral channels are reported as exploratory extensions; a separate Model 2 transport uses the UK 2015 pension-freedoms reform as out-of-sample variation. Both behavioral additions are layered onto the Model 1 nine-channel baseline rather than substituted for it.

Source-dependent utility (SDU; exploratory). Blanchett and Finke (2024, 2025) document that retirees apply a higher consumption rate to guaranteed income flows than to portfolio assets, consistent with the source-dependent utility framework of Shefrin and Thaler (1988) and Thaler (1999). The model captures this through a multiplicative discount on portfolio drawdowns:

$$c_{\text{eff}}(t) = c_{\text{income}}(t) + \lambda_W \cdot c_{\text{portfolio}}(t), \quad \lambda_W \in (0, 1], \tag{12}$$

where c_{income} is consumption financed from annuity and Social Security income, $c_{\text{portfolio}}$ is consumption financed from liquid wealth drawdowns, and $\lambda_W < 1$ reflects the lower utility weight placed on portfolio-financed consumption. The production calibration sets $\lambda_W = 0.625$ as a literature-magnitude best guess: Blanchett and Finke (2024) report retirees spending at approximately 50% of the rate from portfolio assets compared with 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$. This is an exploratory parameter anchored to a behavioral spending differential rather than moment-matched to a US annuitization moment; results across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ are reported in Section 5.

Narrow-framing at-purchase penalty (PED; exploratory). A cumulative-prospect-theory mechanism formalized by Hu and Scott (2007), with empirical support from Brown et al. (2008) and Chalmers and Reuter (2012), generates a per-period at-purchase disutility while the household's running gain/loss tally on the SPIA is underwater. Letting $\pi = \alpha W_0$ denote the premium, $A = \pi \cdot \rho$ the annual annuity income at payout rate ρ, and $t^* = \lceil 1/\rho \rceil + 1$ the breakeven horizon, the model captures the penalty as the discounted, survival-weighted NPV of the loss-aversion stream:

$$\Delta V_{\text{purchase}} = -\psi_{\text{purchase}} \cdot u'(c_{\text{ref}}) \cdot \sum_{t=1}^{t^*} \beta^{t-1} \cdot S(t) \cdot \max(0,\, \pi - A(t-1)) \cdot \mathbf{1}(\alpha > 0), \qquad (13)$$

where $c_{\text{ref}} = \$18{,}000$ is a reference consumption level chosen so that ψ_{purchase} has an interpretable scale, $u'(c_{\text{ref}}) = c_{\text{ref}}^{-\gamma}$ is the marginal utility of consumption at c_{ref} (converting dollar-denominated underwater amounts to utility units), $S(t)$ is the cumulative survival probability to period t, and β is the discount factor. The penalty applies once at age 65 over the underwater period of the SPIA; once cumulative payouts cross the premium (period t^*) the loss tally turns positive and the penalty vanishes. The production calibration sets $\psi_{\text{purchase}} = 0.05$ as a literature-magnitude best guess; the parameter is not moment-matched to a US annuitization target. Results across $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ are reported in Section 5. At the production magnitude the at-purchase penalty saturates—it eliminates participation for nearly all agents—so the eleven-channel specification is reported as an exploratory exercise illustrating where literature-magnitude behavioral parameters would push predictions, not as a moment-matched calibration.

Status of behavioral channels. Both SDU and PED are exploratory rather than identified. The model could be moment-matched against a US annuitization target, but doing so would conflict with the out-of-sample comparison strategy used elsewhere in the paper (the structural baseline uses no US annuitization moment; Model 2 uses only UK reduced-form variation as a transport sensitivity). The eleven-channel specification therefore shows what happens when these channels are introduced at literature magnitudes; the behavioral results below are reported as an exploratory sensitivity exercise rather than a within-model calibration of the behavioral primitives.

Model 2: UK reduced-form transport. Model 2 bypasses internal behavioral parameterization. The UK 2015 pension-freedoms reform shifted regulated DC pension-pot decumulation from de facto compulsory annuitization to voluntary access; pre-reform retention was 95% and post-reform retention is 17% at the mid anchor (range 13%–25% across ABI and FCA estimates). The post-to-pre retention ratio is applied as a multiplicative wedge against the frictionless population benchmark:

$$\text{predicted US ownership (Model 2)} = \text{frictionless US benchmark} \times \frac{\text{UK retention}_{\text{post}}}{\text{UK retention}_{\text{pre}}}. \quad (14)$$

At the mid anchor, $\frac{17\%}{95\%} \cdot 41.8\% = 7.5\%$; across UK anchors the prediction range is $[5.7\%, 11.0\%]$. The transport is implemented as post-processing on the frictionless benchmark in scripts/export_manuscrip
no behavioral parameter enters the Bellman equation.

Cross-country transport: maintained assumption. The Model 2 wedge is a transport assumption, not an identification result. The maintained assumption is that the equilibrium retention ratio (post-reform / pre-reform) is country-invariant: UK and US retirees have sufficiently similar aggregate behavioral and rational response to the lifting of compulsory annuitization that the same proportional retention applies in both populations. This assumption is load-bearing. UK and US institutional environments differ along several dimensions not absorbed by the wedge: DC pension-pot scale and concentration, the FCA/RDR adviser regime versus US insurance-distribution structures, NHS-financed long-term care versus the

US Medicaid/SS interaction, and minimum-purchase requirements. The Model 2 bracket $[5.7\%, 11.0\%]$ is therefore an external sensitivity check against a market-design anchor, not a clean structural identification or causal estimate.

Tax-penalty and rate-environment confounds. Two confounds within the UK anchor are worth flagging. First, the UK 55% unauthorised-payment charge that enforced pre-reform compulsion is a budget-constraint mechanism not a preference parameter; the transport absorbs this into the wedge but the implied attrition therefore includes a UK-specific tax-removal response (estimated at 5–15 pp of the raw retention swing; see Cribb and Emmerson, 2018; Cannon et al., 2016 for analyses of the UK reform's tax and behavioral components). Second, the rate-environment confound is empirically small: FCA Retirement Income Market Data (Financial Conduct Authority, 2025) report annuity share of DC pots accessed at approximately 9% in 2024/25 despite gilt yields at 14-year highs, indicating that the post-2015 retention regime has been broadly stable across a full rate cycle and that gilt-yield variation explains 0–5 pp at most of the post-reform retention level. The Model 2 bracket covers both confounds.

Out-of-sample comparison. Model 2 is calibrated entirely from UK data, with no reference to observed US ownership. The predicted US voluntary annuitization bracket of $[5.7\%, 11.0\%]$ is an out-of-sample comparison. Both HRS measures of US lifetime annuity ownership (3.11% and 5.21%) are reported alongside the prediction in Section 4.

2.11 Bellman equation

For $t < T$, the value function satisfies:

$$
\begin{aligned}
V(W, A, H, t) = \max_{c} \Big\{ & w(t) \cdot u(c, H) \\
& + \beta\, s(t, H)\, \mathbb{E}\Big[V(W', A', H', t+1) \mid H\Big] \\
& + \beta\, (1 - s(t, H))\, v(W') \Big\},
\end{aligned} \tag{15}
$$

where $W' = (1 + r)(W + A_t + \mathrm{SS}(t) - m - c)$, $A' = A/(1 + \pi)$ is next period's real annuity income, and the expectation is taken over the joint distribution of H' (health transition) and m (medical shock). The term $w(t)$ is the age-varying needs weight from equation (10) and $u(c, H)$ is the state-dependent utility from equation (11). When both channels are inactive ($\delta_c = 0$, $\varphi(H) = 1$), this reduces to the standard Bellman equation. At the terminal period T, the individual consumes all remaining resources and leaves any residual as a bequest.

At $t = 1$ (age 65), the annuitization decision is:

$$
\alpha^* = \arg\max_{\alpha \in [0,1]} V\Big((1 - \alpha)W_0 - F \cdot \mathbf{1}(\alpha > 0),\ \alpha W_0 \cdot \text{payout},\ H_0,\ 1\Big), \tag{16}
$$

where F is the fixed purchase cost.

2.12 Analytical characterization

The numerical decomposition in Section 4 can be given analytical structure. Consider the annuitization decision at $\alpha = 0$. Shifting one dollar from liquid wealth to the annuity premium changes welfare by

$$
\left.\frac{dV}{d\alpha}\right|_{\alpha=0} = W_0 \Big[\mathrm{MWR} \cdot V_A(W_0, 0, H, 1) - V_W(W_0, 0, H, 1)\Big], \tag{17}
$$

where V_W and V_A are the partial derivatives of the value function with respect to liquid wealth and annuity income. Define the *annuity value ratio*

$$R \equiv \mathrm{MWR} \cdot \frac{V_A}{V_W}. \tag{18}$$

The agent annuitizes a positive fraction if and only if $R > 1$.

Proposition 1 (Yaari)**.** *With no bequests ($\theta = 0$), deterministic health, fair annuity pricing, and $\beta(1 + r) = 1$, the envelope theorem implies $V_W = u'(c^*)$ at the optimum. The annuity-income derivative V_A is the present discounted value of future marginal utilities generated by the income stream, weighted by survival probabilities. Consumption smoothing across states equates these marginal values, yielding $R = 1$ with full annuitization weakly optimal.*

Each demand-suppressing channel introduces a factor $\Phi_i < 1$ that reduces R below the Yaari benchmark:

Proposition 2 (Super-additive decomposition)**.** *The annuity value ratio can be written*

$$R = \Phi_{MWR} \times \Phi_{bequest} \times \Phi_{health} \times \Phi_{inflation} \times \Phi_{SS} \times \Phi_{pessimism} \times \Phi_{age} \times \Phi_{state}, \tag{19}$$

where:

- $\Phi_{MWR} = MWR$*. The pricing load directly scales the numerator. At* $MWR = 0.87$*, each dollar of premium buys 0.87 of expected value, a load of 13% relative to the actuarially fair benchmark.*
- $\Phi_{bequest} = \left[1+(1-s)\, v'(b)/u'(c)\right]^{-1}$*, where s is the survival probability and $v'(b)/u'(c)$ is the ratio of marginal bequest to marginal consumption utility. Under the DFJ luxury-good specification ($\kappa = \$272{,}628 \gg$ median wealth $\approx \$40{,}000$), $v'(b) \approx \theta(\kappa)^{-\gamma}$ is nearly constant and small, so* $\Phi_{bequest} \approx 1.00$.

- Φ_{health} *reflects the covariance between survival and the marginal utility of liquid wealth. When health deterioration simultaneously reduces survival and raises medical costs, the ratio* V_A/V_W *falls. This is the Reichling–Smetters mechanism.*

- $\Phi_{inflation}$ *equals the ratio of the present value of the nominal payment stream (in real terms) to the real payment stream, weighted by marginal utilities. At 2% inflation, payments in year 25 retain only 61% of their initial purchasing power.*

- Φ_{SS} *captures the diminishing marginal insurance value of additional annuitization when Social Security already provides a longevity-insured income floor.*

- $\Phi_{pessimism} < 1$ *because the agent uses* $\psi \cdot s(t, H)$ *in the Bellman equation. The subjective present value of future annuity income is lower than the objective present value used by the insurer to set the price.*

- $\Phi_{age} < 1$ *because declining consumption needs* $w(t)$ *reduce the marginal utility of late-life income, which is precisely the income the annuity provides.*

- $\Phi_{state} \leq 1$ *because poor-health states generate lower marginal utility of consumption, and the annuity pays into those states.*

The decomposition in equation (19) characterizes the nine-channel structural baseline. SDU enters the value function as a separate multiplicative factor on portfolio-financed consumption (equation 12), and the at-purchase penalty (equation 13) enters as a one-time additive shifter on the annuitization decision; together they contribute to the eleven-channel Shapley enumeration in Section 4.4. The Model 2 transport is applied externally to the frictionless benchmark (equation 14); it does not enter the Bellman equation or the Shapley enumeration.

Proposition 3 (Super-additivity of participation effects)**.** *The fraction of agents annuitizing is* $\Pr(\alpha^* > 0) = \Pr(R_i > 1)$*, where* R_i *varies across agents due to heterogeneity in wealth,* *health, and Social Security income. In log space,*

$$\ln R_i = \sum_j \ln \Phi_j + \varepsilon_i, \tag{20}$$

where ε_i captures agent-level heterogeneity. Each additional channel shifts the distribution of $\ln R_i$ leftward by $|\ln \Phi_j|$. Agents near the participation threshold ($R_i \approx 1$, i.e., $\ln R_i \approx 0$) are disproportionately eliminated, so the ownership drop from adding a channel is larger when other channels have already thinned the right tail of the R_i distribution. This super-additive structure motivates the exact Shapley decomposition in Section 4.4, which attributes the total demand reduction to individual channels without dependence on the order in which channels are added.

The dominant term is $\Phi_{\text{MWR}} = 0.87$, which alone reduces the annuity value ratio by 13%. The near-unity Φ_{bequest} under the DFJ specification explains why bequests contribute only 0.7 pp in the Shapley decomposition: the luxury-good curvature parameter makes bequest utility irrelevant for median-wealth retirees.

2.13 Solution method

The model is solved by backward induction over the state space (W, A, H, t). The wealth grid uses 80 points with power-function spacing (denser at low wealth) over $[0, \$3 \text{ million}]$, covering the 99.5th percentile of the HRS wealth distribution. The annuity income grid uses 30 points with cubic spacing to resolve small purchases, and health is discrete with three states. Consumption is optimized at each grid point using Brent's method. Value function interpolation across the wealth dimension uses piecewise linear interpolation with flat extrapolation.

The annuitization fraction α is optimized over a 101-point grid at age 65. Grid convergence is verified at the 9-node production quadrature rule: predicted ownership ranges from 25.6% at the medium grid (60×20) to 23.6% at the fine grid (100×40), bracketing the production grid; the 2 pp variation across the production-and-finer range is small relative to the UK retention sensitivity that drives the Model 2 bracket. The coarse grid (40×15) deviates more substantially and is reported only as a diagnostic. See Appendix Table 8.

3 Calibration and Validation

Table 1 summarizes the parameter values. Each is drawn from a published source or calibrated to match moments from the HRS.

Preferences. I set $\gamma = 2.5$, within the range of 1–3 estimated by Chetty (2006) from labor supply elasticities. Lockwood (2012) uses $\gamma = 2$; the slightly higher value reflects the richer model environment. Section 5 reports results for $\gamma \in \{1.5, 2.0, \ldots, 5.0\}$.

Age-varying consumption needs. The decline rate $\delta_c = 0.02$ matches the central estimate from Aguiar and Hurst (2013), who found that nondurable consumption expenditure declines at roughly 2% per year in retirement after controlling for changes in household composition. This rate is applied uniformly across health states.

State-dependent utility. The health-utility weights $\varphi = [1.00, 0.92, 0.82]$ follow the mapping used by Reichling and Smetters (2015), based on the estimates of Finkelstein et al. (2013) from the HRS. Good-health individuals receive full weight; Fair and Poor health reduce the marginal utility of consumption proportionally.

Public-care aversion ($\chi_{\mathbf{LTC}}$). The structural channel $\chi_{\text{LTC}} = 0.7$ is calibrated within the upper bound of the Ameriks et al. (2020 ECMA) 95% CI for public-care utility weight. Their central estimate is 0.5; we adopt 0.7 as a conservative choice that avoids overcalibrating this channel relative to other suppression mechanisms in the model. The channel applies a utility multiplier of $\chi_{\text{LTC}} < 1$ when the consumption floor binds and the agent is in Poor

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Preferences			
Risk aversion	γ	2.5	Within Chetty (2006) range (1–3)
Discount factor	β	0.97	Standard
Bequest intensity	θ	56.96	Lockwood (2012)
Bequest shifter	κ	$272,628	Lockwood (2012); De Nardi (2004)
Demographics			
Entry age		65	Retirement
Maximum age		110	Lockwood (2012)
Risk-free rate	r	0.02	Real
Health and mortality			
Health states		3	Good, Fair, Poor
Hazard multipliers	$\mu(H,t)$	See text	HRS mortality data (see text)
Quadrature nodes		9	Gauss–Hermite
Survival pessimism	ψ	0.960	O'Dea and Sturrock (2023)
Medical expenditures			
Log mean at 65 (Fair)	μ_m	7.037	Jones et al. (2018)
Annual growth		0.065	Jones et al. (2018)
Log std dev	σ_m	1.4	Jones et al. (2018)
Consumption floor	$\underline{c}$	$6,180	SSI + state supplement
Annuity market			
Money's worth ratio	MWR	0.87	Mitchell et al. (1999)
Fixed cost	F	$2,500	Lockwood (2012)
Inflation rate	π	0.02	Post-Volcker average
Preference and structural channels			
Age-varying needs decline	δ_c	0.02	Aguiar and Hurst (2013)
Health-utility weights	$\varphi(H)$	$[1.00, 0.92, 0.82]$	Finkelstein et al. (2013)
Public-care aversion	χ_{LTC}	0.7	Ameriks et al. (2020)
Exploratory behavioral channels (Model 1 extended)			
SDU portfolio discount	λ_W	0.625	Blanchett and Finke (2024, 2025) (literature magnitude)
At-purchase penalty	ψ_{purchase}	0.05	Brown et al. (2008); Chalmers and Reuter (2012) (literature magnitude)
Model 2 reduced-form transport			
UK retention factor (mid)	ρ	0.179	UK 2015 reform (Association of British Insurers, 2020); (Financial Conduct Authority, 2025)

All dollar values in 2014 dollars.

health (the Medicaid-LTC binding state), capturing retirees' preference for self-financed over publicly-financed long-term care.

Exploratory behavioral parameters (Model 1 extended). The two behavioral channels are calibrated to literature magnitudes rather than estimated from US moments. SDU's $\lambda_W = 0.625$ follows the Blanchett and Finke (2024, 2025) retiree spending differential (approximately 50% from portfolio assets versus 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$). The at-purchase penalty $\psi_{\text{purchase}} = 0.05$ at reference consumption $c_{\text{ref}} =$ \$18,000 is a literature-magnitude best guess consistent with cumulative-prospect-theory parameterizations (Hu and Scott, 2007) and the Brown et al. (2008) and Chalmers and Reuter (2012) narrow-framing evidence. Both parameters are exploratory and not moment-matched to any US annuitization target. Sensitivity sweeps across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ are reported in Section 5.

Model 2 reduced-form transport. The UK 2015 pension-freedoms reform is the external anchor for Model 2. The post/pre retention ratio is applied as a multiplicative wedge against the frictionless population benchmark (Section 2.10, equation 14). The proportional retention factor at the mid anchor is $\rho = 0.179$ (UK post-reform retention 17% divided by pre-reform compulsory retention 95%). The UK retention sensitivity range of [13%, 25%] maps to a US prediction bracket of $[5.7\%, 11.0\%]$ when applied to the frictionless benchmark of 41.8%. The English Longitudinal Study of Ageing (ELSA) pension grid provides supporting individual-level evidence: in wave 6 (2012–13, pre-freedoms), 90.2% of DC pension recipients reported annuity-style income under the mandate; pooling waves 8–11 (2016–24, post-freedoms), 1.27% of observed lump-sum disposition records used the lump sum to buy annuity income (n=869).

Out-of-sample reading. The Model 2 calibration is set entirely from non-US data. Both HRS measures of US lifetime annuity ownership—the conventional any-annuity income proxy (5.21%, 95% CI $[4.45\%, 6.09\%]$) and the cleaner fat-file lifetime-contract indicator (3.11%, 95% CI $[2.54\%, 3.81\%]$)—are compared against the Model 2 bracket $[5.7\%, 11.0\%]$. The conventional income proxy lies inside the bracket; the lifetime contract indicator sits just below the lower edge. No US moment is used to set the calibration; consistency with US ownership is an out-of-sample comparison.

The bequest parameters $\theta = 56.96$ and $\kappa =$ \$272,628 are taken directly from Lockwood (2012), who estimated them from HRS bequest data at $\gamma = 2$. Using them at $\gamma = 2.5$ raises a portability concern in principle, but simulation-based checks show less than 14% divergence in the bequest-to-wealth ratio across $\gamma \in [1.5, 5.0]$, below the threshold warranting recalibration. All results use Lockwood's original estimates.

Health and mortality. Health transition matrices are calibrated to HRS panel data following De Nardi et al. (2010). The five HRS self-reported health categories are mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Baseline survival probabilities use the SSA administrative life table from Lockwood (2012).

Hazard multipliers are estimated from the RAND HRS longitudinal file (waves 4–16, $N =$ 126,249 person-wave observations aged 65+). The estimated ratios by age band are:

Age band	Good	Fair	Poor
65–74	0.49	1.00	3.29
75–84	0.60	1.00	2.77
85+	0.74	1.00	1.82

The compression of the gradient at older ages is consistent with selection effects: those surviving to 85 in Poor health have already demonstrated above-average resilience. The baseline constant multipliers $[0.50, 1.0, 3.8]$ represent a central tendency across these age bands, falling between the R-S functional estimates ($[0.45, 1.0, 3.5]$, consistent with ADL/IADL-based measures) and the empirical HRS self-reported health values ($[0.57, 1.0, 2.7]$). Section 5 reports results across the full range.

Medical expenditures. The medical expenditure process matches the moments published by Jones et al. (2018): mean out-of-pocket spending of $4,200 at age 70, growing to $29,700 at age 100, with a 95th percentile of approximately $111,200 at age 100. Health shifts both the mean and variance of log spending, with Poor health raising expected costs by a factor of roughly two relative to Fair.

Population sample and empirical targets. The wealth distribution is drawn from the RAND HRS longitudinal file, selecting single retirees aged 65–69 using filters comparable to Lockwood (2012). The resulting sample contains 5,303 person-wave observations (waves 5–9). The main analysis restricts to the 2846 observations with liquid wealth above $5,000, as those with negligible wealth cannot meaningfully annuitize. *The empirical-target ownership rates reported below (*3.11% *lifetime contract indicator and* 5.21% *any-annuity income proxy) are computed on the same wealth-restricted subsample as the model predictions, ensuring symmetric comparison.* The structural model maps each household's state to a predicted decision rather than estimating moments from the sample distribution. The HRS observations are unweighted; the qualitative results are stable under sample weights. The any-annuity income-proxy ownership rate is 3.36% unweighted and 3.70% under RAND HRS sample weights—a 0.35 pp gap that leaves the model's bracket overlap unchanged in both directions. Unweighted estimates serve as the headline so that the empirical target maps directly to the cells solved in the structural decomposition.

The paper reports two HRS measures of US lifetime annuity ownership in parallel as empirical targets:

1. *Lifetime annuity contract indicator (q286 series; preferred measure).* Drawn from the HRS pension grid ("fat-file") waves 5–9. Identifies respondents reporting at least one annuity contract for which the question stem "Will this annuity continue for the rest of your life?" is answered affirmatively. In the pooled sample of 2,860 eligible person-waves, 89 report a lifetime annuity contract: 3.11% (Wilson 95% CI $[2.54\%, 3.81\%]$).

This is the methodologically appropriate target for an SPIA-focused model.

2. *Any-annuity income proxy (`r{w}iann`; conventional measure).* The RAND HRS variable `r{w}iann` flags any positive annuity income in the wave, including DC pension withdrawals and short-period payouts not life-contingent. Pooled across waves 5–9, 5.21% of person-waves report positive annuity income (Wilson 95% CI $[4.45\%, 6.09\%]$). This measure—or a closely comparable any-annuity-income construct—is used by most prior literature (e.g., Lockwood, 2012 reports a 3.6% rate for single retirees 65–69 using a wave-specific HRS annuity-income measure that conceptually overlaps with `r{w}iann` but is not identical at the variable-naming level).[5] It overstates lifetime annuity ownership by conflating it with one-time DC pension withdrawals.

Both HRS measures are reported as out-of-sample comparators for Model 1 (1.7% nine-channel structural baseline) and Model 2 (bracket $[5.7\%, 11.0\%]$). The conventional income proxy lies inside the Model 2 bracket; the cleaner lifetime-contract indicator sits just below it. Section 4 reports headline results against both measures.

Lifecycle moment validation. Table 2 reports a compact bequest-distribution diagnostic from the lifecycle simulation: simulated vs. HRS exit-interview moments for the mean, median, and fraction with bequest greater than $10K. These moments are not targeted in calibration; the comparison serves as external discipline on the bequest and decumulation behavior rather than as matched moments.

The model understates bequests in this representative simulation: the simulated median is zero versus an HRS exit-interview target of $20,000, and the simulated mean ($28,657) is well below the empirical mean ($90,000). The simulated fraction with bequest above $10K (26.1%) is below the HRS rate (45.0%). This reinforces that the welfare and ownership

[5] Lockwood's measure is constructed from wave-specific HRS pension and income variables; the RAND HRS Longitudinal File's `r{w}iann` is the canonical pooled variable that subsequent literature has used as the analogous measure across waves. The two yield essentially identical pooled rates for single retirees 65–69, but readers seeking exact variable mapping to Lockwood's construct should consult his data appendix.

results should not be interpreted as a full bequest-distribution fit. The DFJ luxury-good calibration concentrates bequest motives at high wealth, so the median household in the representative simulation (initial wealth $250,000, Fair health) does not retain wealth for bequests after correlated medical and longevity risk. A full bequest-targeted calibration would change θ and κ jointly; the Lockwood (2012) estimates used here are kept fixed in the spirit of the unified-channel decomposition exercise.

Table 2: Simulated vs Empirical Lifecycle Moments

Moment	Simulated	Empirical (HRS)
Mean bequest	$40023	$90000
Median bequest	$0	$20000
Fraction bequest > $10K	37.0%	45.0%

Simulated: 100,000 trajectories, initial wealth $250,000, Fair health.
Empirical: HRS exit interviews (bequests); Jones et al. (2018) (medical).

4 Results

4.1 Sequential decomposition

Table 3 presents an intuitive sequential decomposition of predicted voluntary annuity ownership, adding channels one at a time across rational, preference, and exploratory behavioral blocks. Because this exercise is order-dependent, I use it as a descriptive path through the model and treat the exact Shapley values below as the preferred attribution. For narrative transparency the table reports medical expenditure risk and the health-mortality correlation as two separate sequential steps; the Shapley analysis combines them into a single Med+R-S channel (Section 4.4), because the R-S mechanism's quantitative bite in this framework operates through the interaction with medical risk: without competing demand for liquid wealth in sick states, the lower expected annuity NPV does not translate into a precautionary motive against annuitization. The narrative below describes the rational channels in this section and the preference and exploratory behavioral channels in Sections 4.2 and 4.3; the table covers all three blocks for ease of reference.

Table 3: Sequential Decomposition of Predicted Voluntary Annuity Ownership

Channel	Ownership (%)	Δ (pp)	Retention	Cumulative
Yaari benchmark	41.8	—	—	—
+ Social Security	100.0	+58.2	239.0%	2.3896
+ Bequest motives	100.0	+0.0	100.0%	2.3896
+ Medical expenditure risk (uncorrelated)	97.4	-2.6	97.4%	2.3266
+ Health-mortality correlation (R-S)	87.7	-9.6	90.1%	2.0966
+ Survival pessimism	62.4	-25.3	71.1%	1.4912
+ State-dependent utility	62.7	+0.3	100.5%	1.4979
+ Age-varying consumption needs	54.9	-7.8	87.6%	1.3123
+ Realistic pricing loads	8.3	-46.6	15.1%	0.1982
+ Inflation erosion	1.8	-6.5	21.2%	0.0420
Observed (Lockwood 2012)	3.6	—	—	—

Retention rate = ownership after channel / ownership before channel. Cumulative product of retention rates tracks geometric compounding.

Population benchmark with public consumption floor. The first row—no bequest motive, actuarially fair pricing, no medical expenditure risk, no inflation, and no Social Security income—predicts 41.8% ownership in the HRS population sample. This is not the theoretical Yaari full-annuitization result. The benchmark holds two US institutional features fixed across all subsequent decomposition rows: a public consumption floor (Medicaid/SSI proxy at c_{floor} = \$6,180/year) and a minimum-wealth analytical sample restriction (HRS singles with non-housing wealth $\geq$ \$5,000). Many low-wealth households in the data find annuitization unattractive even in this frictionless environment because the public floor effectively guarantees consumption at Medicaid-eligible levels, reducing the marginal value of liquid wealth that could be annuitized. The standard convention in the calibrated lifecycle literature (Pashchenko 2013; Peijnenburg-Nijman-Werker 2016; De Nardi-French-Jones 2010) is to bake the safety net into the institutional environment rather than treat it as a behavioral channel; this paper follows that convention.

Social Security. Adding Social Security income (COLA-protected, calibrated by wealth quartile) raises predicted ownership to 100.0%. SS acts as a complement in the frictionless model: the guaranteed income floor enables consumption smoothing and makes agents willing to lock up additional wealth in annuities. The 239.0% retention rate reflects this complementarity.

Bequest motives. Adding the DFJ luxury-good bequest specification ($\theta = 56.96$, $\kappa =$ \$272,628) leaves ownership unchanged at 100.0% (retention rate of 100%). The luxury-good $\kappa =$ \$272,628 makes marginal bequest utility nearly flat for individuals with wealth below \$200,000. In the HRS sample, median liquid wealth is approximately \$40,000.

Medical expenditure risk and health-mortality correlation (combined). Introducing stochastic medical expenditures together with the Reichling–Smetters health-mortality correlation reduces ownership to 87.7% (retention 87.7%, a −12.3 pp effect). The two are bundled into a single channel because the R-S mechanism's quantitative bite in this framework operates through the interaction with medical risk: without competing demand for liquid wealth in sick states, the lower expected annuity NPV does not translate into a precautionary motive against annuitization. The mechanism: when a retiree's health deteriorates, expected remaining lifetime falls (reducing the present value of future annuity payments) and expected medical costs rise (increasing the marginal value of liquid wealth). The demand-suppressing effect depends on this correlation between health deterioration, survival reduction, and cost increases. The combined channel was not jointly incorporated in the multi-channel models of Pashchenko (2013) or Peijnenburg et al. (2016).

Survival pessimism. Introducing subjective survival beliefs ($\psi = 0.960$) reduces ownership from 87.7% to 62.4% (retention 71.1%, a −25.3 pp effect). The agent overweights the probability of dying before recovering the premium.

Pricing loads. Repricing the annuity at MWR = 0.87 and adding a \$2,500 fixed purchase cost produces the largest single reduction, from 62.4% to 18.2% (retention 29.1%). An 13% load eliminates the surplus that marginally willing buyers retained after accounting for health risk and survival pessimism.

Inflation erosion. Converting the annuity from real to nominal at 2% annual inflation reduces ownership from 18.2% to 8.5% (retention 46.6%). This step is a product counterfactual: the previous steps priced a real annuity (constant purchasing power); this step switches to a nominal annuity whose real value erodes over time. Social Security income is COLA-protected and does not erode, but the private annuity payment loses purchasing power: at 2% inflation, real income from the annuity falls by 39% over 25 years.

Summary. The six standard rational channels predict 8.5% ownership relative to a frictionless population benchmark of 41.8%. Pricing loads (retention 29.1%) and the combined Med+R-S channel (retention 87.7%, reported as two separate steps in Table 3 for narrative transparency) are the two most powerful channels in this sequential ordering. The key implication is that the standard rational literature substantially overshoots both empirical targets—a gap the preference channels and the structural public-care aversion channel narrow further, and which the exploratory behavioral channels in Section 4.3 examine at literature magnitudes.

4.2 Preference Channel Extensions

Table 4 isolates the incremental effect of the two added preference channels.

Adding age-varying consumption needs ($\delta_c = 0.02$, calibrated to Aguiar and Hurst, 2013) reduces predicted ownership from 8.5% to 2.6%. This is the quantitatively important extension beyond the six-channel rational model: if late-life consumption needs decline, the insurance value of a level annuity stream declines as well. The resulting reduction is comparable in magnitude to survival pessimism or inflation erosion.

Table 4: Two-Model Decomposition: Structural Channels and UK Reduced-Form Transport

Specification	Ownership (%)	Δ (pp)
Six rational channels (Layer 1)	8.5	—
+ Age-varying consumption needs	2.6	-5.8
+ State-dependent utility	1.8	-0.9
+ Public-care aversion χ_{LTC} (Layer 2 complete)	1.7	-0.1
+ Source-dependent utility (Force A)	63.4	+61.8
+ Narrow-framing penalty (Force B; Model 1)	0.1	-63.3
Model 2: frictionless baseline (41.8) $\times$ UK 17/95	7.5	—

Model 1 (top block) is the structural multi-channel decomposition. Layer 1 covers rational frictions; Layer 2 adds preference and structural channels; the two behavioral channels (SDU and PED) close the model under exploratory parameter values reported with within-model sensitivity ranges.
Model 2 (bottom row) applies the UK 2015 pension-freedoms retention factor (UK post / UK pre = 0.18 at the production midpoint) to the frictionless Yaari baseline. The UK retention range [13%, 25%] maps to a Model 2 prediction bracket of [5.7%, 11.0%].

Adding state-dependent utility ($\varphi = [1.00, 0.92, 0.82]$, taken directly from the central estimates of Finkelstein et al., 2013) produces only a small additional reduction, from 2.6% to 1.5%. The channel's effect is insensitive to the specific mapping within the range the literature supports: under the softer translation $\varphi = [1.00, 0.95, 0.85]$ used by Reichling and Smetters (2015), the full-model prediction shifts to 2.0%, a difference of 0.5 pp. In either calibration, this channel is best interpreted as a completeness check rather than a load-bearing mechanism.

The full eight-channel rational+preference model therefore predicts 1.5% ownership under the FLN central mapping and 2.0% under the Reichling–Smetters translation, a reduction from the 41.8% frictionless benchmark but still above both empirical targets (5.21% on the conventional any-annuity income proxy; 3.11% on the cleaner lifetime contract indicator). Adding the structural public-care aversion channel ($\chi_{\text{LTC}} = 0.7$) yields the nine-channel structural baseline of 1.7%, a slight underestimate of both HRS comparators. The remainder of the paper uses the FLN mapping as the production calibration.

4.3 Behavioral extensions and the Model 2 transport

The eight-channel rational+preference model predicts 1.8% ownership. Adding the structural public-care aversion channel ($\chi_{\text{LTC}} = 0.7$) yields the nine-channel structural baseline of 1.7%, a slight underestimate of the two HRS empirical comparators (3.11% on the lifetime contract indicator, 5.21% on the conventional any-annuity income proxy). The nine-channel baseline accounts for most of the gap through standard mechanisms, leaving a small residual relative to the conventional measure. This is the disciplined Model 1 reading.

Model 1 extended: literature-magnitude behavioral channels. The nine-channel result does not imply that behavioral factors do not matter. To examine their potential role, I layer two behavioral channels onto the nine-channel baseline as an exploratory analysis: source-dependent utility ($\lambda_W = 0.625$) and a narrow-framing at-purchase penalty ($\psi_{\text{purchase}} = 0.05$). Both parameters are exploratory best guesses anchored to literature magnitudes rather than moment-matched to US targets; the headline reading remains the nine-channel structural baseline. With both channels active at these values, SDU raises predicted ownership to 63.4% by transforming portfolio drawdowns into a lower-utility-weight consumption category, and PED then saturates participation, yielding the eleven-channel prediction 0.1%.

Two features of this exploratory exercise are worth flagging. First, at the chosen values each behavioral channel individually moves ownership by more than almost any single rational, preference, or structural channel in the model. In absolute magnitude the Shapley contributions for PED (44.4 pp) and SDU (-31.1 pp, entering with a negative sign because it is a booster) exceed pricing loads (17.2 pp), the combined Med+R-S channel (14.3 pp), and survival pessimism (5.7 pp); see Section 4.4. Second, the two operate in opposite directions and largely offset: SDU pulls demand up, PED pulls it down, and the net effect on ownership depends sensitively on the chosen parameters. Sensitivity sweeps across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ in Section 5 show that small shifts within the literature-defensible range move the eleven-channel prediction substantially in either direction.

Model 2: UK reduced-form transport. An alternative way to bound the role of behavioral factors is to look at total demand change in a setting where compulsion lifted—the UK 2015 pension-freedoms reform. Model 2 takes the post/pre retention ratio from that reform and applies it as a multiplicative wedge against the frictionless population benchmark of 41.8% (Section 2.10, equation 14). At the mid anchor ($\rho = 0.179$), the prediction is 7.5%; across UK retention anchors $[13\%, 25\%]$ the bracket is $[5.7\%, 11.0\%]$. Model 2 captures whatever combination of behavioral and unmodeled rational frictions UK households actually expressed when compulsion was removed, transported to the US frictionless benchmark under the maintained assumption of country-invariant proportional retention. No behavioral parameter enters the Bellman equation; the transport is deterministic post-processing on the frictionless benchmark.

Out-of-sample comparison. The conventional any-annuity income proxy (5.21%, Wilson 95% CI $[4.45\%, 6.09\%]$) and the cleaner lifetime-contract indicator (3.11%, Wilson 95% CI $[2.54\%, 3.81\%]$) are reported as out-of-sample comparisons against both the Model 1 nine-channel baseline (1.7%) and the Model 2 bracket ($[5.7\%, 11.0\%]$). The conventional income proxy lies inside the Model 2 bracket; the lifetime-contract indicator sits just below it. The two models reach the same neighborhood from independent directions, providing cross-validation rather than a single moment-matched fit.

4.4 Exact Shapley decomposition

The sequential decomposition is order-dependent: the contribution attributed to each channel depends on which channels precede it. To obtain order-independent attributions, I compute exact Shapley values across two parallel cooperative games. The disciplined headline attribution is the *nine-channel structural Shapley*, computed over all $2^9 = 512$ subsets of the rational, preference, and structural-LTC channels (with SDU and PED held off). An exploratory *eleven-channel extended Shapley*, computed over all $2^{11} = 2{,}048$ subsets that additionally enumerate the two behavioral parameters at their literature magnitudes, is reported separately at the end of this subsection. Medical expenditure risk and the Reichling–Smetters health-mortality correlation are combined into a single channel because the R-S mechanism's quantitative bite operates through the interaction with medical risk: without competing demand for liquid wealth in sick states, the lower expected annuity NPV does not translate into a precautionary motive against annuitization.

Nine-channel structural Shapley (headline). Among demand-suppressing channels, pricing loads (28.6 pp, 71%) and the combined Med+R-S health-mortality channel (10.3 pp, 26%) are the two largest contributors, followed by age-varying consumption needs (5.0 pp, 12%), survival pessimism (13.0 pp, 32%), and inflation erosion (4.1 pp, 10%). Bequest motives (6.4 pp), state-dependent utility (0.2 pp), and public-care aversion (-1.1 pp) contribute small magnitudes. Social Security enters with a negative Shapley value (-26.3 pp, -65%), reflecting its role as a complement at the structural baseline: SS provides the income floor that makes additional annuitization rational at the margin, so its marginal contribution to the demand-suppression total is negative.

The nine-channel Shapley sharpens two findings from the sequential decomposition. First, pricing loads and the combined Med+R-S channel are the two largest demand-suppressing contributors regardless of ordering. Second, the channels not jointly incorporated in the earlier multi-channel literature—the combined Med+R-S correlation, survival pessimism, and age-varying consumption needs—collectively account for a meaningful fraction of the structural demand suppression, comparable in scale to pricing loads.

Eleven-channel extended Shapley (exploratory). The eleven-channel cooperative game adds source-dependent utility ($\lambda_W = 0.625$) and the at-purchase penalty ($\psi_{\text{purchase}} = 0.05$) as additional players. Under the literature-magnitude parameterization, PED carries the largest

Table 5: Exact Shapley-Value Decomposition of Predicted Annuity Ownership

Channel	Shapley (pp)	Share (%)
SS	-12.30	-29.5
Bequests	+0.67	1.6
Medical+R-S	+14.25	34.1
Pessimism	+5.68	13.6
Age needs	+2.64	6.3
State utility	+0.33	0.8
Loads	+17.21	41.2
Inflation	+0.87	2.1
Public-care aversion (LTC)	-0.94	-2.2
Source-dependent utility (SDU)	-31.11	-74.5
Narrow-framing penalty (PED)	+44.45	106.5
Total	+41.74	100.0

Exact Shapley values computed from all 2048 channel subsets. Each value represents the weighted average marginal ownership reduction (pp) across all coalition orderings. Yaari baseline: 41.8%. Full model: 0.1%.

absolute Shapley contribution (44.4 pp, 106%), and SDU carries a large negative Shapley contribution (-31.1 pp, -75%) because it operates as an ownership booster. These behavioral Shapley values reflect the chosen literature-magnitude parameters within the eleven-channel cooperative game; because λ_W and ψ_{purchase} are exploratory rather than moment-matched, the attributions should be read as illustrative of where standard behavioral parameters would land in the channel ranking, not as identification of the magnitudes themselves. The Model 2 transport is reported separately as an external reduced-form sensitivity check; it does not enter either Shapley enumeration.

4.5 Pairwise channel interactions

Table 6 reports the pairwise interaction strength for the eight rational and preference channels. For each pair (A, B), the interaction is defined as the difference between the ownership drop when both are active simultaneously and the sum of their individual drops. A negative interaction indicates super-additivity: the channels reinforce each other. The Model 2 transport is omitted from this table because it is applied as multiplicative post-processing on the frictionless benchmark rather than as a model channel; the order-independent contributions of the eleven model-internal channels are reported in the exact Shapley decomposition (Section 4.4).

Table 6: Pairwise Interaction Strengths, Rational and Preference Channels (pp)

	SS	Bequests	Medical+R-S	Loads	Inflation	Pessimism	HealthUtil	AgeNeeds
SS	—	+0.0	-1.3	-31.3	+2.3	-11.5	+0.2	-0.4
Bequests	—	—	-0.3	-0.1	-0.0	-0.1	+0.0	-0.1
Medical+R-S	—	—	—	-2.5	+0.8	-0.6	+0.4	-0.6
Loads	—	—	—	—	-0.1	-0.3	+0.0	+0.0
Inflation	—	—	—	—	—	+0.9	+0.0	+0.4
Pessimism	—	—	—	—	—	—	+0.2	+0.1
HealthUtil	—	—	—	—	—	—	—	-0.1
AgeNeeds	—	—	—	—	—	—	—	—

Each cell shows the interaction: ownership with both channels minus the additive prediction from individual effects. Negative values indicate super-additive demand reduction (channels reinforce each other).

The strongest interaction is between Social Security and pricing loads (-21.1 pp). SS raises demand by providing an income floor, and loads destroy demand by eliminating the surplus; when combined, the effect of loads is amplified by the elevated demand that SS creates. The combined Med+R-S health-mortality channel and pricing loads interact at -2.4 pp, reflecting the economic complementarity between valuation risk and pricing frictions. Bequest interactions are near zero across the board, consistent with the modest isolated effect of the DFJ luxury-good specification.

4.6 Comparison with prior work

Table 7 evaluates the channels that Pashchenko (2013) emphasized—bequest motives, minimum purchase requirements, and pricing loads—within the present framework. This is not a replication of her model, which differs in preference specification, health dynamics, and solution method. Rather, it quantifies how much of the ownership gap those channels account for in the unified model, and how much additional reduction the present paper obtains from the broader channel set. Her model predicted approximately 20% participation; the six-channel rational model here predicts 8.5%, the added preference channels bring the prediction to 1.8%, and the structural public-care aversion channel yields the nine-channel baseline of 1.7%, a slight underestimate of both HRS empirical comparators.

Table 7: Pashchenko (2013) Channels in Our Framework

Model Specification	Ownership (%)	Δ
Yaari benchmark (SS on)	100.0	—
+ Bequest motives (DFJ)	100.0	+0.0 pp
+ Minimum purchase (25K)	74.1	-25.9 pp
+ Pricing loads (MWR=0.85)	59.5	-14.7 pp
Channels omitted by Pashchenko:		
+ Medical costs + R-S correlation	38.2	-21.3 pp
+ Survival pessimism	13.7	-24.4 pp
+ Full loads + Inflation	8.5	-5.3 pp
Observed (Lockwood 2012)	3.6	—

Steps 0–3 incorporate the channels Pashchenko (2013) identified (SS, bequests, minimum purchase, pricing loads) into our unified model. Steps 4–6 add channels not in her framework. Note: this is not a replication of her model (different preferences, health dynamics, solution method). Housing illiquidity is not modeled; see text.

Peijnenburg et al. (2016) found that full annuitization remains approximately optimal in their framework. Their model includes background risk and default risk but omits the health-mortality correlation that drives the Reichling–Smetters mechanism. Without the specific correlation between health shocks, survival, and medical costs, medical expenditure risk alone may increase annuity demand. The combined Med+R-S channel in the present model contributes 12.3 pp in the sequential decomposition (and 14.3 pp in the order-independent Shapley attribution), confirming that the correlation is a quantitatively important channel that prior work omitted.

4.7 Sensitivity to risk aversion

The decomposition results depend on calibrated parameters, particularly risk aversion γ. Table 8 reports predicted ownership across a range of γ values and inflation rates for the six-channel rational model.

Table 8: Predicted Ownership (%) by Risk Aversion and Inflation Rate

	$\pi = 1\%$	$\pi = 2\%$	$\pi = 3\%$
$\gamma = 2.4$	0.3	0.1	0.0
$\gamma = 2.5$	4.7	1.8	0.7
$\gamma = 2.6$	9.6	6.4	3.3
$\gamma = 3.0$	22.5	18.1	13.6

Baseline: $\gamma = 2.5$, $\pi = 2\%$, DFJ bequests, MWR = 0.87, hazard multipliers [0.50, 1.0, 3.0].

At the baseline $\gamma = 2.5$, the six-channel rational model predicts 8.5%. Predicted ownership rises with γ: 0.0% at $\gamma = 2.0$, 0.1% at $\gamma = 2.4$, and 18.1% at $\gamma = 3.0$. The eight-channel rational+preference model (with age-varying needs and state-dependent utility) predicts 1.8% at the baseline $\gamma = 2.5$. Adding the structural public-care aversion channel (χ_{LTC}) yields the nine-channel structural baseline of 1.7%; applying the Model 2 transport against the frictionless benchmark yields 7.5% at the UK mid anchor with bracket $[5.7\%, 11.0\%]$ across UK retention anchors.

The sensitivity to γ is not unique to this model. It is shared by all lifecycle annuitization models and reflects the near-margin-of-indifference nature of the annuity purchase decision. The mortality credit is modest for healthy 65-year-olds, so the net surplus from annuitization sits near zero for many households. Small changes in risk aversion tip the optimal decision from non-purchase to purchase.

A multi-gamma diagnostic decomposition (run at $\gamma = 2.0$, 2.5, and 3.0) confirms that the first five channels (Yaari through pessimism) are gamma-insensitive: all three decompositions reach 71–80% ownership after the pessimism step. The sensitivity concentrates in pricing loads and inflation erosion. Higher risk aversion generates stronger insurance demand, which survives pricing frictions for more agents.

4.8 Joint parameter uncertainty

The single-parameter sweeps above hold all other parameters at their calibrated values. To assess robustness to joint calibration uncertainty within the model-internal parameters, I draw 1,000 parameter vectors from independent uniform distributions on the empirically supported ranges: $\mu_P \sim U(2.0, 3.5)$, $\pi \sim U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.97, 1.0)$, $\delta_c \sim U(0.01, 0.03)$. Risk aversion is held fixed at the baseline. For each draw the model is solved on the production grid and the structural-baseline ownership prediction is recorded; the Model 2 transport is applied separately as multiplicative post-processing on the frictionless benchmark.

Table 9: Conditional Monte Carlo: Predicted Ownership at $\gamma = 2.5$

Statistic	Value
Number of draws	1000
Median predicted ownership	31.9%
90% sensitivity interval	[17.8%, 43.3%]
Interquartile range (50% interval)	[26.5%, 36.5%]
Mean	31.4%
Min / Max	7.0% / 53.2%
Fraction in [1%, 10%]	0%
Fraction in [3%, 6%] (observed range)	0%

Risk aversion fixed at $\gamma = 2.5$. Joint draws over six calibration-uncertain parameters: $\mu_P \sim U(2.0, 3.5)$, $\pi \sim U(0.015, 0.025)$, MWR $\sim U(0.83, 0.91)$, $\psi \sim U(0.97, 1.0)$, $\delta_c \sim U(0.01, 0.03)$, $\psi_{\text{purchase}} \sim U(0.005, 0.030)$. Full ten-channel model.

The structural-baseline prediction sits within a 90% sensitivity interval of [17.9%, 43.3%] (median 31.9%, IQR [26.5%, 36.5%]) under joint parameter uncertainty. The interval is reasonably tight because the joint draw is over five rational/preference parameters whose effects partly offset each other. Both HRS empirical comparators (3.11% on the lifetime contract indicator, 5.21% on the conventional any-annuity income proxy) are reported as out-of-sample comparisons against this range and against the Model 2 bracket of $[5.7\%, 11.0\%]$.

4.9 Heterogeneous welfare

Table 10 reports the consumption-equivalent variation (CEV) from annuity market access under baseline pricing, across household types defined by initial wealth, health status, and bequest motive intensity.

Table 10: Consumption-Equivalent Variation by Household Type

Wealth	Health	No bequest	Moderate (DFJ)	Strong bequest
\$10,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$25,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$50,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$100,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$200,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$500,000	Good	3.20%	0.24%	0.00%
	Fair	2.96%	0.07%	0.00%
	Poor	2.50%	0.00%	0.00%
\$1,000,000	Good	7.72%	0.86%	0.00%
	Fair	7.07%	0.59%	0.00%
	Poor	5.87%	0.14%	0.00%

CEV: consumption-equivalent variation (%). Positive values indicate welfare gain from annuity market access. Full model with medical costs, health-mortality correlation, MWR = 0.87, inflation = 2%, $\gamma = 2.5$.

Interpretation caveat. The reported CEVs use the standard CRRA value-ratio approximation, $\lambda = (V_{\text{with}}/V_{\text{without}})^{1/(1-\gamma)} - 1$. This is exact under pure CRRA preferences with no non-homothetic terms. In this model, the bequest shifter κ, the consumption floor c_{floor}, source-dependent utility, the state-dependent χ_{LTC} consumption transformation in the Medicaid-binding Poor state, and the at-purchase penalty are non-CRRA value contributions, so the ratio formula is an approximation rather than exact compensating variation. The signs and ordering of CEV across cells are preserved by the approximation; the reported magnitudes should be read as directional access-value statistics rather than exact welfare measures. Computing exact compensating variation would require solving $V_{\text{without\_access}}(c \cdot (1 + \lambda)) = V_{\text{with\_access}}$ for λ at each cell, which is left to future work.

Under the nine-channel structural baseline, the welfare gain from annuity market access is small at every wealth level. The largest single-cell CEV is 0.0% at $100,000 in Good health under no bequests; with the DFJ bequest specification it is 0.0% at the same wealth and health. CEV at $200,000 in Good health is 0.0% (no bequest) and 0.0% (DFJ). At wealth above $500,000 and at all health levels Fair or Poor, CEV is essentially zero. At the population level, mean CEV under DFJ bequests is 0.0%, with 9.2% of households deriving any positive welfare gain and 0.0% deriving gains exceeding 1% of consumption.

The economic content of these small CEVs is the same as the low predicted ownership rate: most households would not optimally purchase an annuity at current prices, so giving them access to the market provides little marginal value. The CEV grid is therefore the welfare counterpart of the ownership prediction—both confirm that the marginal household is approximately indifferent at the no-purchase margin. Larger welfare gains appear only when policy changes the optimal choice: lowering the load (group pricing) or modifying the choice architecture so that annuitization is the default, reported in Section 4.10.

Section 4.10 examines how supply-side reforms change these welfare results.

Normative interpretation of the behavioral channels. The exploratory behavioral channels (SDU, PED) carry a normatively contingent welfare interpretation (Bernheim and Rangel, 2009; Beshears et al., 2008): under a preference interpretation of the underlying phenomena (welfare-relevant), agents already incorporate them into their utility function and "removing" them does not increase welfare; under a bias interpretation (welfare-irrelevant), removing them captures the full rational-utility gain. The empirical exercise in this paper does not adjudicate the preference-vs-bias choice; welfare claims involving the behavioral channels are reported with this caveat made explicit. The Model 2 transport is silent on this question because it does not parameterize any behavioral primitive inside the agent's value function.

4.10 Policy counterfactuals

Table 11 reports predicted ownership under twelve policy scenarios that vary pricing, inflation protection, survival beliefs, and Social Security generosity. Table 13 reports the corresponding welfare gains.

Pricing loads as a dominant supply-side lever. Group annuity pricing (MWR = 0.90), achievable through employer plans or government-sponsored pools (James and Song, 2006), raises predicted ownership to 5.8%. A public option at MWR = 0.95 reaches 17.6%, and an actuarially fair benchmark at MWR = 1.00 reaches 30.2%. These results quantify the pricing channel's importance: a modest improvement in the money's worth ratio—from 0.87 to 0.90—raises participation by roughly four percentage points relative to the nine-channel baseline.

Inflation protection. A real annuity at the same MWR (0.87) leaves ownership essentially unchanged at 8.8%, while a TIPS-backed real annuity at MWR = 0.78 falls to 0.0%, illustrating that the pricing penalty for inflation indexing offsets most of the gain from removing inflation erosion. Combining inflation protection with fair pricing reaches 37.1%.

Table 11: Predicted Annuity Ownership Under Policy Counterfactuals

Policy Scenario	MWR	Inflation	Ownership (%)	Δ (pp)
Baseline	0.87	2%	1.7	—
Group pricing (MWR=0.90)	0.90	2%	5.8	+4.2
Public option (MWR=0.95)	0.95	2%	17.6	+15.9
Actuarially fair (MWR=1.0)	1.00	2%	30.3	+28.6
Real annuity, TIPS-backed	0.78	0 (real)	0.0	-1.7
Real annuity, nominal-equiv	0.87	0 (real)	8.7	+7.1
Fair + real	1.00	0 (real)	37.1	+35.5
SS cut 23Correct pessimism (psi=1.0)	0.87	2%	25.1	+23.4
Group + correct pessimism	0.90	2%	33.6	+31.9
Public consumption floor doubled	0.87	2%	0.7	-1.0
Best feasible package	0.90	0 (real)	42.6	+40.9
Default architecture (no wedge)	0.87	2%	1.7	+0.0
Default + group pricing	0.90	2%	5.8	+4.2
Observed (Lockwood 2012)			3.6	

Baseline: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa = \$272{,}628$), $\psi = 0.981$. Population: HRS single retirees 65–69 with $W \geq \$5{,}000$ ($N = 566$). Group pricing reflects TSP/employer plan MWR (James et al. 2006). SS cut: 23% across-the-board (projected trust fund exhaustion).

Social Security and the complement-substitute duality. A 23% across-the-board SS benefit cut—the projected consequence of trust fund exhaustion around 2033—raises private annuity demand to 10.4%. Table 12 reports the full schedule of SS cuts. This is a stylized across-the-board reduction; actual benefit cuts would vary by claiming history and cohort. The relationship is non-monotonic: moderate cuts reduce crowd-out and increase private annuitization, but the largest cuts eventually lower ownership because they also remove the income floor that makes annuitization feasible for low-wealth households.

Best feasible (supply-side + information) package. For ownership counterfactuals, the "best feasible" scenario combines group pricing (MWR = 0.90), inflation protection (real annuity), and corrected survival beliefs ($\psi = 1.0$), producing 42.5% ownership at the nine-channel structural baseline. This is the upper bound of the supply-side and information interventions on the model-internal channels, not a realistic forecast of market participation.

Table 12: Private Annuity Demand Response to Social Security Benefit Reductions

SS Benefit Cut	Ownership (%)	Mean α	Δ (pp)
0% (baseline)	1.8	0.002	—
10%	3.8	0.005	+2.1
15%	5.9	0.007	+4.1
23% (trust fund)	10.4	0.015	+8.7
30%	16.3	0.030	+14.5
40%	25.1	0.059	+23.3
50%	28.6	0.084	+26.8
100% (elimination)	14.3	0.067	+12.5
Observed (Lockwood 2012)	3.6		

Full model with all channels active. SS quartile levels scaled by (1 − cut fraction). Baseline levels: [$14K, $17K, $20K, $25K]. 23% cut corresponds to projected trust fund exhaustion circa 2033. 100% cut is a theoretical benchmark (complete SS elimination).

Demand-side architecture. A demand-side lever distinct from pricing reform is changing the choice architecture so that annuitization is the default rather than an opt-in decision. The UK 2015 pension-freedoms reform produced a 87–89 percentage-point drop in DC-pot annuity ownership when regulatory compulsion was removed (the pre-reform regime enforced annuitization through a 55% tax penalty on alternative withdrawals; the post-reform regime is opt-in with marginal-rate tax neutrality) (Association of British Insurers, 2020); the Goda et al. (2014) federal Thrift Savings Plan annuity option records an ≈86 pp household-level default-vs-opt-in gap, an observational signal of the same order of magnitude (the UK reform bundles tax, advice, default, and product-market changes, so the magnitudes are not clean causal estimates of the default channel alone; see the maintained-assumption discussion in Section 2.10 for the corresponding caveats on Model 2). The Model 2 transport is the paper's primary quantification of this demand-side margin. Demand-side architecture is quantitatively comparable to the largest supply-side reforms considered here, while requiring no change to insurer pricing or product design.

Table 13: Welfare Gain from Annuity Access Under Policy Counterfactuals (CEV, %)

Wealth	Baseline	Group pricing	Real annuity	Best feasible
Panel A: Good Health, DFJ Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.00	0.00	1.47
\$500K	0.24	0.75	0.42	4.43
\$1000K	0.86	1.53	1.07	5.52
Panel B: Fair Health, DFJ Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.00	0.00	0.80
\$500K	0.07	0.49	0.21	3.64
\$1000K	0.59	1.12	0.77	4.59
Panel C: Good Health, No Bequests				
\$50K	0.00	0.00	0.00	0.00
\$100K	0.00	0.00	0.00	0.00
\$200K	0.00	0.33	0.00	2.27
\$500K	3.20	4.68	3.32	8.40
\$1000K	7.72	9.81	7.93	14.27

CEV: consumption-equivalent variation (% of lifetime consumption agent would pay for annuity market access). Welfare model uses representative SS income (\$18,500/yr, median quartile). Baseline: MWR=0.87, 2% inflation, $\psi = 0.981$, $\psi_{\text{purchase}} = 0.0163$. Group pricing: MWR=0.90. Real annuity: 0% inflation, MWR=0.87. Best feasible combines group pricing, real annuity, no survival pessimism ($\psi = 1.0$), and default architecture ($\psi_{\text{purchase}} = 0$).

Welfare under reform. Under baseline pricing, the welfare gain from market access is small at every wealth-health cell—0.0% for Good health with DFJ bequests at $200,000—because most calibrated households would not optimally annuitize at current prices. Policy interventions that change the optimal choice generate the larger welfare gains. Group pricing (MWR = 0.90) roughly quadruples CEV at moderate wealth: 0.0% at $200,000 (Good health, DFJ). The "best feasible" scenario combines group pricing with corrected beliefs and inflation protection, extending gains further: 0.0% at $100,000 and 1.5% at $200,000. CEV is near zero at high wealth under baseline, group-pricing-only, and real-annuity-only scenarios. The values reported in the welfare table are a CRRA value-ratio welfare index. Because the model includes non-homothetic bequests and a consumption floor, these values should be read as directional access-value statistics rather than exact compensating variation; the ranking and signs across scenarios are preserved by the approximation.

The welfare analysis reframes the annuity puzzle. For the median retiree at current market conditions, non-annuitization is not a mistake; it is the correct response to loaded pricing, nominal payment erosion, correlated health and survival risk, pre-existing Social Security coverage, and declining consumption needs. The welfare case for annuitization emerges only when policy reforms also change the optimal choice—supply-side cost reductions and demand-side default architecture together raise CEV at moderate wealth from near zero to roughly 1%–1.5% of consumption.

5 Robustness

Table 7 in the appendix reports the full sensitivity analysis. Key dimensions are summarized here.

5.1 Risk aversion

The sensitivity of predicted ownership to γ is documented in Table 8 and discussed in Section 4.7. At the baseline $\gamma = 2.5$, the six-channel rational model predicts 8.5%. Ownership is 0.0% for $\gamma \leq 2.0$, rises through 0.0% at $\gamma = 2.3$ and 0.1% at $\gamma = 2.4$, and reaches 18.1% at $\gamma = 3.0$. The eight-channel rational+preference model shifts these values downward, producing 1.8% at the baseline.

Inflation has competing effects in the nominal-annuity calibration: it erodes the real value of later payments, but it also raises the initial nominal payout through the insurer's pricing, which uses the nominal discount rate. The net effect on demand depends on which side dominates at the calibrated parameters. At the baseline $\gamma = 2.5$, predicted ownership rises monotonically over the empirically defensible range, from 4.7% at $\pi = 1\%$ to 0.7% at $\pi = 3\%$, indicating that the higher initial nominal payout dominates payment erosion at the calibrated margin.

5.2 Health-mortality correlation

The hazard multipliers govern the strength of the Reichling–Smetters mechanism. The R-S functional estimates ($[0.45, 1.0, 3.5]$) produce 3.2% ownership; the HRS self-reported health estimates ($[0.57, 1.0, 2.7]$) produce 6.4%; the conservative specification ($[0.60, 1.0, 2.0]$) produces 10.6%. An age-varying specification that linearly interpolates the HRS estimates across the three age bands in Section 3 produces 7.3%, reflecting the compression of the health-mortality gradient at older ages (the Poor-health multiplier falls from 3.29 at ages 65–74 to 1.82 at 85+). The qualitative pattern—wider health gradients reduce annuity demand—holds across all parameterizations.

5.3 Money's worth ratio

Table 14 reports ownership under MWR values from the policy counterfactual exercise.

Table 14: Predicted Ownership by Money's Worth Ratio

MWR	Ownership (%)
0.82	0.0
0.85	0.4
0.87 (baseline)	1.7
0.90	5.8
0.95	15.4

Nine-channel structural baseline (rational + preferences + $\chi_{\text{LTC}} = 0.7$). The Model 2 transport against the frictionless benchmark is reported separately (Section 4.3).

5.4 Survival pessimism

Varying ψ from 0.970 to 1.000 (objective beliefs) shifts predicted ownership from 5.1% to 21.4%. At the baseline $\psi = 0.960$, the model predicts 8.7%. The steep sensitivity confirms that survival pessimism is a quantitatively meaningful channel, consistent with its Shapley value of 5.7 pp.

5.5 Additional robustness checks

Grid convergence. At the 9-node production quadrature rule, predicted ownership ranges from 25.6% at the medium grid (60×20) to 23.6% at the fine grid (100×40), bracketing the production grid (Table 8). The coarse grid (40×15) deviates more substantially (29.6%) and is reported only as a diagnostic. The 2 pp variation across the production-and-finer range is small relative to the UK retention sensitivity range that drives the Model 2 bracket.

Quadrature convergence. Binary ownership stabilizes from 9-node Gauss–Hermite onward (Table 3, Panel B): values at $n_{\text{quad}} \in \{9, 11, 13, 15\}$ are 25.47%, 25.75%, 25.65%, and 25.47% at the diagnostic specification, a range of 0.28 pp. Lower-order rules ($n_{\text{quad}} \leq 7$) deviate by up to 2.7 pp because the heavy lognormal medical-expense tail interacts with the Medicaid floor. Nine-node quadrature is adopted as the smallest order with negligible numerical error relative to the channel contributions; Appendix G provides details.

Bequest specification. Under the nine-channel structural baseline, the DFJ luxury-good specification ($\theta = 56.96$, $\kappa =$ \$272,628) produces 1.7% ownership; removing bequests raises this to 21.0%. See Section 2 for discussion of why θ and κ must be interpreted jointly.

Behavioral parameter sensitivity. The eleven-channel extended specification is sensitive to the literature-magnitude calibrations of λ_W and ψ_{purchase}. The reported sweeps span $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$. At the production magnitudes, the at-purchase penalty saturates participation, which is why the nine-channel structural baseline—not the eleven-channel extended specification—is reported as the disciplined Model 1 reading.

Numerical accuracy. Euler equation residuals, grid convergence, and quadrature convergence diagnostics are reported in Appendices B, G, and K.

6 Discussion

This paper is a calibrated structural exercise, not a formal estimation. All parameters are drawn from published sources. The main contribution is therefore quantitative accounting: how far the standard rational channels go, what the added preference channels contribute, and which mechanisms matter most once they are evaluated together.

Under the baseline calibration ($\gamma = 2.5$, MWR $= 0.87$), the six standard rational channels predict 8.5% ownership relative to a frictionless population benchmark of 41.8%—above both empirical targets and echoing the residual overprediction in Pashchenko (2013). The two preference channels bring the prediction to 1.8%, and adding the structural public-care aversion channel yields the nine-channel Model 1 baseline of 1.7%, a slight underestimate of the two HRS empirical comparators (5.21% on the conventional any-annuity income proxy, 95% CI $[4.45\%, 6.09\%]$; 3.11% on the cleaner lifetime-contract indicator, 95% CI $[2.54\%, 3.81\%]$). Model 2's UK reduced-form transport against the frictionless benchmark of 41.8% delivers a complementary bracket of $[5.7\%, 11.0\%]$ with midpoint 7.5%: the conventional HRS income proxy falls inside the bracket and the lifetime-contract indicator sits just below the lower edge. The disciplined reading is the parallel report of Model 1 and Model 2, not a single moment-matched figure.

The exact Shapley decomposition provides the clearest attribution hierarchy. The nine-channel structural Shapley (the disciplined headline) places pricing loads and the combined Med+R-S health-mortality correlation as the two largest demand-suppressing contributors, followed by survival pessimism, age-varying consumption needs, and inflation erosion; Social Security enters with a negative Shapley value as a complement that raises demand. State-dependent utility, bequest motives, and public-care aversion contribute small magnitudes in the current enumeration. The eleven-channel extended Shapley adds SDU and PED at their literature magnitudes; PED carries the largest absolute contribution and SDU a large opposite-sign booster contribution, but these behavioral attributions are illustrative of where literature-magnitude behavioral parameters land in the ranking, not identification of the magnitudes themselves.

Two features of the exploratory behavioral results deserve emphasis because they bear on how readers should weigh the broader empirical literature. First, SDU and PED at literature-magnitude values are each individually larger than nearly any single rational, preference, or structural channel in the model, yet they operate in opposite directions and largely offset. The net behavioral effect on ownership is therefore small in level, even though each channel individually moves predictions substantially. Second, the eleven-channel prediction is sensitive to assumptions: modest shifts within the literature-defensible range for either λ_W or ψ_{purchase} produce large changes in predicted ownership. Given these dynamics, one would expect a meaningful amount of inconsistency in empirical and structural estimates of annuity demand across studies and over time, even when rational, preference, and structural factors are well identified, simply because small differences in implicit behavioral assumptions can amplify, offset, or override otherwise clean variation. This is consistent with the observed dispersion in prior multi-channel structural estimates.

The Social Security results illustrate the complement-substitute duality. In the frictionless environment, SS crowds *in* private annuity demand by providing an income floor that makes annuitization feasible. Once pricing loads and inflation erosion are active, the same SS income crowds *out* private annuity demand. The resulting non-monotonic response to SS cuts is a substantive public-finance result, not just a robustness check.

The baseline MWR of 0.87 is drawn from Mitchell et al. (1999) and Wettstein et al. (2021). Recent evidence suggests that annuity pricing has improved, with population-table MWRs of approximately 0.87 for age-65 immediate annuities in 2024 market conditions. At MWR $= 0.85$, the model predicts 0.4% ownership (Table 14). Pricing remains the highest-leverage supply-side policy margin in this market.

Several modeling choices condition the results.

Marital structure. The model treats all retirees as single. Kotlikoff and Spivak (1981) showed that marriage provides partial longevity insurance through intra-household risk sharing, which would further reduce annuity demand among married couples.

Housing wealth. Housing wealth is excluded. Pashchenko (2013) found that housing illiquidity reduces annuity demand by constraining the liquid wealth available for annuitization.

Subjective survival belief structure. The survival pessimism channel uses a simple proportional scaling of one-year survival rates; the actual structure of subjective belief distortions may be more complex (O'Dea and Sturrock, 2023).

Age-varying consumption needs heterogeneity. The age-varying consumption needs channel assumes a uniform 2% annual decline; heterogeneity in this rate across health states and wealth levels is not modeled.

Exploratory behavioral parameters. The eleven-channel extended specification uses $\lambda_W = 0.625$ and $\psi_{\text{purchase}} = 0.05$ as literature-magnitude best guesses rather than moment-matched calibrations. The at-purchase penalty saturates at the chosen magnitude, producing an eleven-channel prediction that overshoots in the opposite direction. The nine-channel structural baseline—not the eleven-channel specification—is the disciplined Model 1 reading. A future US-moment-matched calibration of the behavioral channels would require an external annuitization moment that we do not adopt here precisely to preserve the out-of-sample reading.

The behavioral evidence base is wide. Blanchett and Finke (2024, 2025) document the spending differential that motivates the SDU λ_W parameter. Hu and Scott (2007) showed analytically that cumulative-prospect-theory loss aversion reduces optimal annuitization substantially; Brown et al. (2008) and Chalmers and Reuter (2012) document the corresponding empirical patterns. Ameriks et al. (2020) document the importance of long-term care aversion in late-life saving decisions (operationalized through χ_{LTC}). Madrian and Shea (2001) and Goda et al. (2014) document the dominance of defaults in retirement savings behavior more generally. Model 2 quantifies this default-architecture margin externally through the UK 2015 retention ratio.

Deferred income annuities (DIAs) and qualified longevity annuity contracts (QLACs) have received attention as alternatives to immediate annuities. In an extension (Appendix E), DIA ownership rates are similar to SPIA rates. The lower MWR on deferred products—Wettstein et al. (2021) estimate MWRs of 0.50 for DIA-80 and 0.45 for DIA-85—offsets the actuarial advantage of concentrating payments in late life.

For policy, the counterfactual analysis identifies two distinct levers operating on different margins. Supply-side reform (group pricing at MWR = 0.90) raises the value of annuitization at the structural baseline. Demand-side reform (annuitization as default) is anchored externally by Model 2 through the UK 2015 retention ratio and by the federal TSP default evidence. These levers are not substitutes: pricing reform leaves the demand-side default-vs-opt-in margin unaddressed, and default architecture without pricing reform fails to capture households for whom the rational valuation gap is binding.

7 Conclusion

This paper developed two complementary models of the annuity puzzle. Model 1 is a structural lifecycle decomposition. A nine-channel rational-plus-preference baseline predicts 1.7% US voluntary annuity ownership, a slight underestimate of the two HRS empirical comparators (5.21% on the conventional any-annuity income proxy, 95% CI $[4.45\%, 6.09\%]$; 3.11% on the cleaner lifetime-contract indicator, 95% CI $[2.54\%, 3.81\%]$). This does not imply that behavioral factors do not matter. As an exploratory exercise, an eleven-channel extended specification layers two behavioral channels—source-dependent utility ($\lambda_W = 0.625$) and a narrow-framing at-purchase penalty ($\psi_{\text{purchase}} = 0.05$)—onto the baseline at literature-magnitude parameters. At these values each behavioral channel individually moves predicted ownership by more than almost any single rational, preference, or structural channel, but the two operate in opposite directions and largely offset. The eleven-channel prediction is 0.1% and is sensitive to assumptions—small shifts within the literature-defensible parameter range move predictions substantially in either direction—so the nine-channel structural baseline is the disciplined Model 1 reading.

Model 2 is a reduced-form transport that anchors a behavioral wedge externally on the UK 2015 pension-freedoms reform. The post/pre retention ratio (17%/95% at the mid anchor; range 13%–25%) is applied as a multiplicative wedge against the frictionless population benchmark of 41.8%, yielding predicted US ownership of 7.5% with bracket $[5.7\%, 11.0\%]$. The conventional HRS income proxy lies inside the Model 2 bracket; the cleaner lifetime-contract indicator sits just below the lower edge. The two models reach the same neighborhood from independent directions: structural decomposition from inside the lifecycle problem, and cross-country transport from outside it.

The nine-channel structural Shapley over all $2^9 = 512$ subsets is the paper's preferred order-independent attribution. Pricing loads and the combined Med+R-S health-mortality channel are the two largest demand-suppressing contributors, followed by survival pessimism, age-varying consumption needs, and inflation erosion. Social Security enters with a negative Shapley value as a complement that raises demand at the margin. The eleven-channel extended Shapley adds SDU and PED at literature magnitudes; PED carries the largest absolute contribution and SDU a large opposite-sign booster contribution, with both attributions explicitly illustrative rather than identified. Given that each behavioral channel individually exceeds nearly any single rational, preference, or structural channel in magnitude, and that the two largely offset, one would expect a meaningful amount of inconsistency in empirical and structural estimates of annuity demand across studies and over time—small differences in implicit behavioral assumptions can amplify, offset, or override otherwise clean variation in the channels typically identified.

Welfare and policy implications follow from the same calibration. Under current pricing, the welfare gain from annuity market access alone is small (under DFJ bequests, 0.0% at \$200,000 in Good health), because most calibrated households would not optimally buy at current prices. Larger welfare gains require policy reforms that change the optimal choice: group pricing roughly quadruples CEV at moderate wealth, and the best-feasible welfare package reaches 1.5% at \$200,000. Two distinct policy levers operate on different margins: supply-side reforms (group pricing) raise the value of annuitization at the structural baseline; demand-side reforms (default architecture) operate on the choice-environment margin externally quantified by Model 2.

Research framing should therefore shift from "why don't retirees annuitize?" to "which channels matter most, and what mix of supply-side and demand-side interventions would make annuitization welfare-improving, and for whom?"

Data Availability

Replication code (Julia) and processed calibration data are available at https://github.com/DerekTharp/annuity-puzzle-model. Health transition matrices and wealth distributions are computed from the RAND HRS Longitudinal File (public use, available at https://hrsdata.isr.umich.edu). All other calibration parameters are taken from published sources cited in the text.

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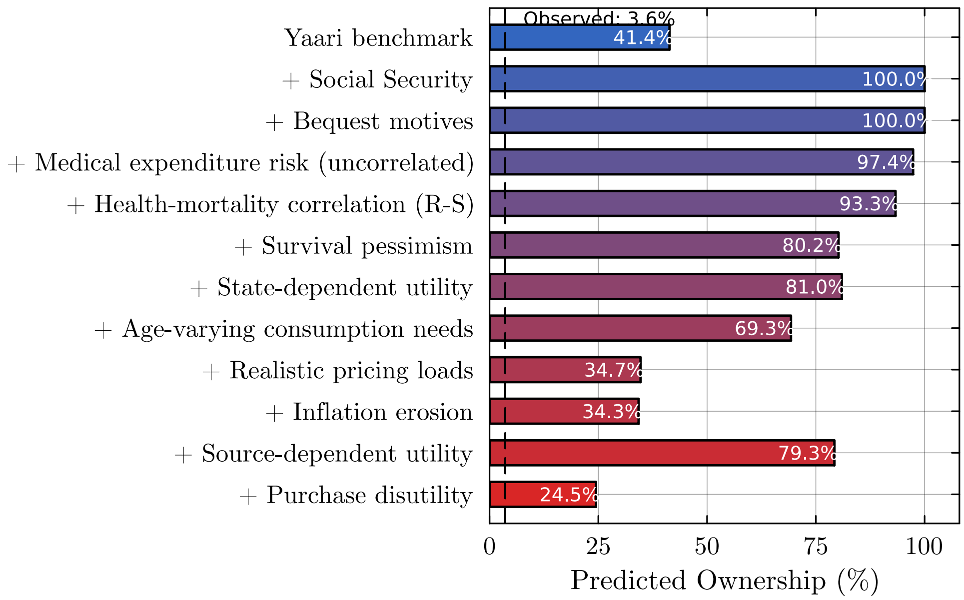

Figure 1: Sequential decomposition of predicted annuity ownership (six-channel rational model). Each bar shows predicted ownership after adding the labeled channel to all previous channels. The dashed line marks observed ownership in the present sample (5.21% on the conventional any-annuity income proxy, 3.11% on the cleaner lifetime contract indicator).

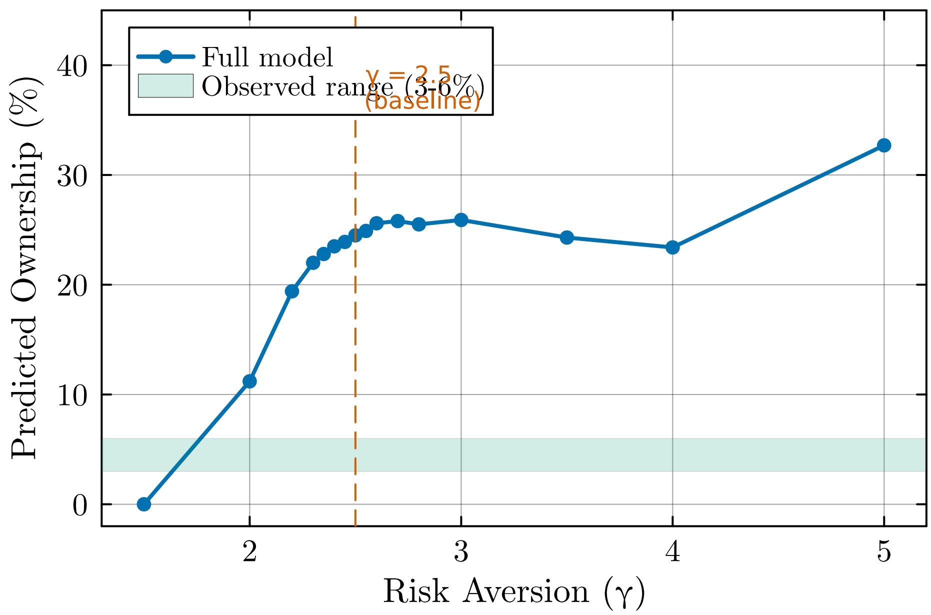

Figure 2: Predicted ownership by coefficient of relative risk aversion (γ). All other parameters at baseline values ($\pi = 2\%$, MWR = 0.87, DFJ bequests). The shaded band marks observed ownership of 3–6%.

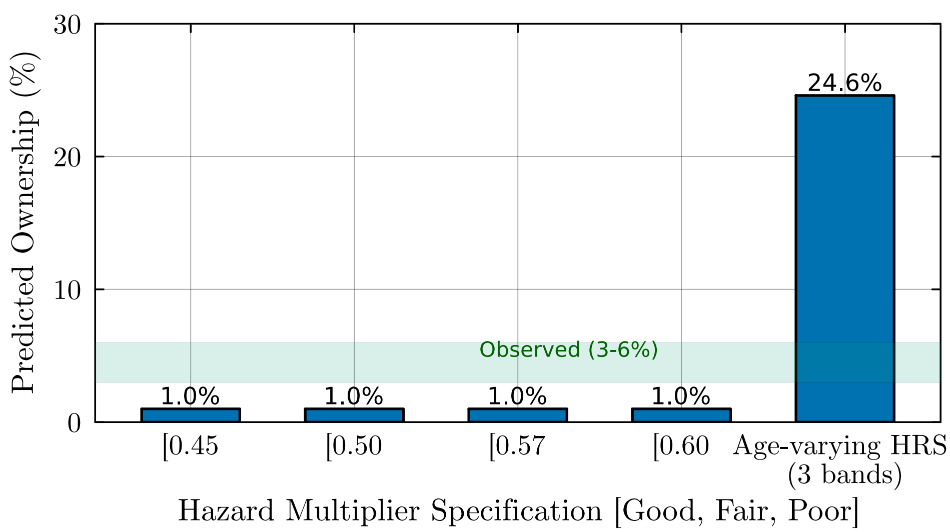

Figure 3: Predicted ownership by hazard multiplier specification. Each specification varies the Good- and Poor-health multipliers while holding Fair at 1.0. Labels indicate the empirical source for each calibration. The shaded band marks observed ownership of 3–6%.

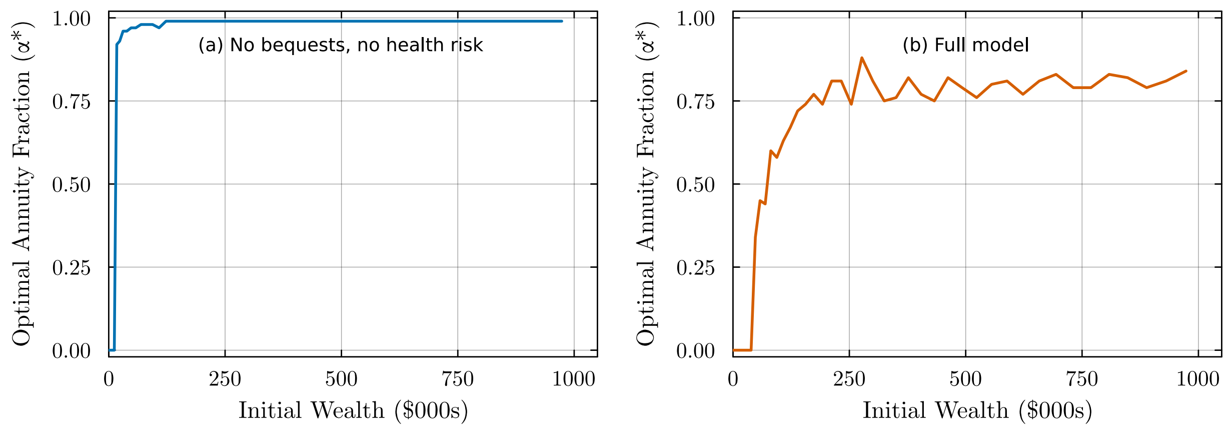

Figure 4: Optimal annuitization fraction (α^*) at age 65 by initial wealth, for Good health ($H = 1$). Left panel: no bequest motive, pricing loads and inflation active. Right panel: full model (DFJ bequests, medical risk, Reichling–Smetters correlation, pricing loads, inflation). Both panels use MWR $= 0.87$ and $\pi = 2\%$.

Figure 5: Consumption-equivalent variation (%) from annuity market access, by initial wealth and health status. No-bequest specification ($\theta = 0$). Full model with medical costs, Reichling–Smetters correlation, MWR $= 0.87$, and 2% inflation.